

# SESTINA v1.0: Exact Inference, Robust Evaluation, and Adversarial Testing in Six-Player Canadian Fish

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## Abstract

Six-player Canadian Fish is a decentralised imperfect-information team game in which every action is public, so the hidden state reduces to the initial deal and the posterior over that deal can be computed exactly. We develop and evaluate SESTINA v1.0, the strongest configuration produced in this project’s FishBot lineage. It combines an approximate Sinkhorn fit to that posterior, started from a fitted policy prior, with a linear ask and declaration policy, a public-history tie-breaking rule that preserves common knowledge among teammates, a half-suit contestation weighting, a deduction-state stall detector in place of an event-count termination rule, and a guarded determinized test-time search. Evaluation follows a protocol registered in advance and run on sealed holdout material, using duplicate deal blocks, deal-clustered bootstrap confidence intervals, replication across two disjoint deal banks as an advance-specified criterion, calibrated detection floors, and mechanical side-channel controls.

Against **F-cheap**, the cheapest configuration genuinely on the v0.6 frontier and the registered comparison target, SESTINA v1.0 achieves a +3.33 percentage-point win-rate edge (95% CI [+2.88, +3.78]) over 48,000 sealed games, with the sign replicating on both banks. The advantage persists under cross-play between independently trained runs and eight rule dialects. Under partner substitution against a v0.5 opponent 6 of 7 changed-partner rows stay positive, but the worst is  $-0.19$  [ $-1.04, +0.67$ ], which that battery does not resolve against the registered  $-1.00$  collapse threshold. Separately, none of eight independently constructed adversarial searches found a positive edge at the tested budgets.

SESTINA v1.0 does not, however, measurably outperform a composite configuration assembled earlier in the same programme (+0.15 pp, 95% CI [ $-0.29, +0.59$ ]), indicating that the architecture work which followed added no measurable strength. Over a shared 31-member opponent panel SESTINA v1.0’s worst cell is  $-0.04$  pp [ $-1.41, +1.33$ ], which does not replicate in sign, and it is 3rd of four on minimax regret. Four candidate mechanisms failed to produce measurable improvement at this resolution. At the calibrated resolution of this evaluation the tested policy class appears locally flat, and we state what evidence a near-optimality claim would require.

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# 1 Introduction

Six-player Canadian Fish, also called Literature, is a team card game with an unusual combination of properties. Every action is public, so the entire hidden state is the initial deal and the posterior over that deal is exactly computable. The unit of play is a three-player team whose members cannot communicate. And because a deployed team consists of three identical deterministic agents, its coordination is common knowledge, which inverts the usual assumption in the team-correlation literature that a team’s correlation device is private. The result is a domain in which inference can be made provably correct, so that any remaining loss is attributable to policy rather than to belief.

Computational work on the six-player game appears to be extremely limited. In the searches conducted across this project’s cycles we found no academic work on it, no public repository containing search, planning or a working learner for it, and no coverage in RLCard [9] (§3.1). The nearest published agent plays a four-player, 48-card variant that does not raise the six-player allocation problem.

This paper presents **SESTINA v1.0**, the strongest agent produced in this project’s FishBot lineage, and reports a rigorous evaluation of it. Earlier versions established the engine, the exact inference machinery and a sequence of progressively stronger policies, but their development also exposed serious evaluation problems: comparisons that could not resolve a one-point effect, an optimiser that could not move a policy, and at least two published conclusions that turned out to be artefacts of the measurement rather than properties of the agent (§4). SESTINA v1.0 therefore asks a question that only becomes answerable once those failures are corrected: whether further measurable progress in this policy class is available at all.

**A note on the name.** SESTINA v1.0 was developed and evaluated as **FishBot v0.7**, the seventh cycle of the lineage, and was renamed at release; its predecessors keep the names they were published under. Every seed, deal bank, digest, directory and configuration string in the evidence base still carries a v07 identifier, including the frozen configuration v07:r12=25, . . . of §6 and the sealed banks of §7. Those names are load-bearing: the banks were committed under them, the freeze artifact round-trips through them, and the digests in Appendix G are digests of objects named that way. Renaming them after the fact would break the provenance chain this paper’s claims rest on, so they are left exactly as they were sealed. Throughout, *SESTINA v1.0* names the agent and *the v0.7 cycle* names the programme that produced and measured it. The name changed at release; nothing measured did.

## 1.1 Research questions

**RQ1** Does SESTINA v1.0 achieve a replicated improvement over **F-cheap**, the cheapest configuration genuinely on the v0.6 frontier and the target registered in advance, measured on material sealed before SESTINA v1.0 existed?

**RQ2** Does that improvement survive changes of partner, cross-play against independently trained policies, alternative rule dialects, and adversarial responses fitted against it?

**RQ3** Which of SESTINA v1.0’s changes account for the improvement, and which candidate mechanisms fail to produce measurable strength?

**RQ4** What do the results imply about the headroom remaining in the current policy class?

§11 answers RQ1 and RQ2 directly, §9 and §11.4 answer RQ3, and §12 takes up RQ4.

## 1.2 Three senses of “strongest”, kept separate throughout

This paper makes a comparative claim of a specific and limited kind, and the distinction matters enough to state before any number appears.

**Lineage strength.** SESTINA v1.0 is the strongest configuration this project has produced, measured against its own predecessors and against the frontier of configurations available at the end of the v0.6 cycle. This is what the paper establishes.

**Robustness.** Whether that strength persists against a fixed opponent panel, under substituted partners, across rule dialects, and against adversaries fitted to exploit it. This the paper evaluates extensively, and reports mixed results.

**Global standing.** Whether SESTINA v1.0 is the strongest Canadian Fish agent anyone has built. **This the paper does not establish and does not claim.** The comparison class contains only agents this project wrote, and every exploitability figure here is a lower bound produced by a search rather than a bound on what a search could find.

Accordingly the paper never describes SESTINA v1.0 as near-optimal, unexploitable, or the strongest known Fish-playing team. §12.7 states what evidence those claims would require, none of which currently exists.

## 1.3 Contributions

1. **SESTINA v1.0** (§6), the strongest configuration in the FishBot lineage, described at the level of its inference, ask and declaration policy, coordination assumptions, termination rule and test-time search.
2. **An exact observer-conditioned deal inference method** for six-player Canadian Fish (§5), which factorises the constraint system by half-suit and joins the blocks through a dynamic program over per-seat card counts, yielding the normalisation constant, per-card marginals and named-allocation probabilities without sampling.
3. **An evaluation framework** (§7) built from duplicate deal blocks, deal-clustered bootstrap intervals, power computed before a cell is run, replication across disjoint banks as an advance-specified criterion, sealed holdout material with committed digests, a soundness gate applied before any strength measurement, calibrated detection floors, and mechanical side-channel controls with planted positive instances.
4. **A comprehensive evaluation of SESTINA v1.0** (§11) covering frontier comparisons, attribution of the gain to components, partner substitution, cross-play, rule dialects, adversarial search including a white-box transcript-inversion adversary, negative controls, and computational cost.
5. **Negative and diagnostic results** (§9.4, §12.3) showing that four plausible architectural mechanisms fail to produce measurable improvement, and identifying a resolution barrier: none of the investigated architectural mechanisms produced a replicated gain above the instrument’s effective resolution at the budgets used here, and the adversary-search detection floor does not fall as evaluation games increase, so exploitability is limited by search power rather than sample size.

## 2 Six-player Canadian Fish

This section states the game completely, so that nothing later depends on outside material, and then says why the domain is scientifically interesting. Readers who know the game should still read

§2.2, because published descriptions differ on several consequential points and the rule set studied here is one explicit selection among them.

## 2.1 Rules

Six players form two teams of three, seated alternately so that no two teammates are adjacent. The deck is 54 cards: the standard 52 plus two jokers. It is partitioned into nine **half-suits** of six cards each, being for each suit the low half 2 through 7 and the high half 9 through A, plus a ninth half-suit made of the four eights and the two jokers. Every card belongs to exactly one half-suit, and each player is dealt nine. The team that takes at least five of the nine half-suits wins.

Play proceeds by **asks**. On their turn a player names one opponent and one card. The ask is legal only if the named card lies in a half-suit that has not yet been claimed, the asker holds at least one *other* card of that half-suit, the asker does *not* hold the named card, and the target is an opponent who still holds at least one card. If the target holds the card they must surrender it and the asker keeps the turn; otherwise the turn passes to the target. The requirement that an asker already hold a card of the half-suit is what makes the game a game, because it turns every ask into a public declaration of partial ownership.

A half-suit is claimed by a **declaration**. A declaring player names, for each of the six cards of a half-suit, which member of their *own* team holds it. The claim is all-or-nothing: if every assignment is correct the declaring team scores the half-suit, and if any assignment is wrong the opposing team scores it. A declaration is legal only while the half-suit is unclaimed, and a seat with no legal ask left is compelled to declare, including a half-suit it holds entirely.

Everything is public. Every ask, its answer, every card transfer, every declaration and its outcome, and every player's card count are observed by all six players. Nothing is hidden except the initial deal.

## 2.2 The rule dialect studied

Published descriptions of this game document mutually incompatible variants [1], so the rule set below is a selection rather than a canon. It matches the strategy corpus this line of work draws on [2] and the rules of record supplied by the project owner. Six of the eight contested choices Appendix A lists are engine switches, so their effect is measurable rather than assumed, and §11.7 measures the principal comparison over eight dialect rows: the default and seven variations. The misdeclaration award is hardcoded and was not varied at all; the forced-endgame ladder has no flag of its own inside the scored harness and moves only inside the `--legacy` bundle, which is why §11.7 can price it only as a residual. The consequential choices are a 54-card deck in nine half-suits with nine cards per player; declaration legal at any moment, in or out of turn, by a seat holding cards or not; a misdeclaration awarding the half-suit to the opposing team; lowest seat index first among simultaneous declarers; an eight-level willingness ladder in the **forced endgame** that arises when one whole team is cardless; and a 400-ask action cap whose residue is adjudicated neutrally by physical majority. No prearranged signalling conventions are available to any policy, and team communication is limited to the forced-endgame willingness bit and to the successor a cardless turn-holder names. Appendix A lists each choice against the switch that changes it.

Two rules are places where the engine is deliberately *more* restrictive than the rules of record allow, and both close coordination channels a policy could otherwise exploit. Both were measured rather than argued about. When the turn reaches a cardless seat whose teammates both hold cards, the rules permit open negotiation over who receives it; the engine instead has the cardless seat choose unilaterally. That channel is empty: a genuine multi-candidate transfer decision arises



0.148 times per game, and replacing the unilateral rule with a *ground-truth* chooser on one team, paired on common deals, is worth  $-0.0007 \pm 0.0024$  half-suits per game over 6,000 paired games. A legal mechanism cannot beat the clairvoyant one, so that interval bounds the whole channel. Similarly, the engine arbitrates simultaneous declarations by lowest seat index rather than by stated confidence, because a confidence comparison would move a statistic of one seat’s hand to five others. That refusal costs 0.37 pp of win rate pooled over 30,000 games, and a clairvoyant arbitrator is worth only +0.35 pp on the same design.

### 2.3 Why the domain is scientifically interesting

**The hidden state is a single object, and its posterior is exact.** Because a card changes hands only through an observed ask, the only thing any player does not know is the initial deal, and the public record determines every subsequent card position given that deal. In many hidden-information games the object that must be filtered is the set of reachable *histories*, for which constructing even one consistent history is FNP-complete in the worst case, with efficient enumeration available exactly when the public tree is sparse [23]. Fish escapes the question entirely: there is exactly one object to reconstruct, and the constraints on it are logical rather than probabilistic. A player told “I do not hold the seven of hearts” knows it, permanently and with certainty.

The research consequence is that Fish permits inference and policy to be *separated*. In poker or bridge, a claim that a policy is weak can always be answered with the suggestion that its beliefs are wrong. Here the beliefs can be made provably correct, at which point every remaining loss is attributable to the policy. That separation is what makes this paper’s negative results interpretable rather than merely discouraging. It also sets up the most counterintuitive finding in the lineage, reported in §4.3: making the belief exact makes the policy worse.

**The team is the unit, and the team cannot talk.** Three teammates share a payoff, hold different private information, and have no communication channel beyond their public actions. This is the decentralised control problem the common-information approach formalises, in which a notional coordinator who knows the common information issues *prescriptions* mapping each controller’s private information to an action [52]. In Fish the common information is the entire public history, so the coordinator’s information state is exactly the belief object the engine already maintains. The coordinator’s problem is not solvable here, since a prescription maps every possible nine-card hand to an ask, so the reduction serves as a correctness lens rather than an algorithm. What it does yield cheaply is a property the cooperative-search literature has to work for [28]: any *deterministic* procedure reading only public state and its own hand is automatically common knowledge among the three teammates, with no shared seed and no communication. The determinism qualifier is essential rather than incidental, and §6.2 is where it becomes an engine switch.

**A published deterministic team inverts a standard assumption.** The team-correlation literature uniformly assumes that a team’s correlation device is private and unknown to the opponent [53, 54, 55]. A team of three identical, deterministic, published agents inverts this: the device is common knowledge and an adversary may read it. §8 takes that seriously as a threat model, and §11.8 reports what an adversary that reads the target’s policy and inverts its transcript actually achieves. In the literature and repositories we reviewed we found no prior work covering this case.

**The domain is cheap, which is a methodological asset.** A deal resolves in roughly a hundred public events, and on one fifteen-thread desktop the engine plays a six-seat mirror of the blueprint policy at 326 complete games per second, with the deployed belief running in every seat; the

evaluated configuration, which searches, costs at least  $4.52\times$  the non-searching blueprint (§11.10). This is what makes the evaluation programme of §7 possible: 48,000 paired games for a single headline comparison, two disjoint holdout banks for every claim, a 31-member opponent panel scored identically for four configurations, and eight independent adversarial searches all run on one desktop. In a domain where one evaluation cell costs GPU-hours, the version of this study that reports its own negative results could not have been afforded. The corresponding liability appears in §12.5: cheap evaluation is not external evaluation.

**What the domain does not offer.** Two disclaimers bear on how later results should be read. First, Fish’s perfect-information game is degenerate rather than merely fused. A clairvoyant player names only cards the target holds, so never fails an ask, never loses the turn, and declares each half-suit as soon as their team holds all of it; almost every reachable position carries the same perfect-information value. Determinization, the standard recipe for hidden-hand card games [11, 12], therefore collapses here to choosing the ask most likely to hit, blind to the information an ask leaks, the information a refusal supplies, and the option value of postponing a declaration. §3.2 places this against the literature. Second, Fish has no passive continuation: a player must ask or declare, so there is no equivalent of the check-and-call rollout on which the standard cheap exploitability probe is built [38]. §12.7 names this as the largest single obstacle between this work and a defensible near-optimality claim.

### 3 Related work

This section covers the work this study builds on and the classes of work it deliberately declines. Absence is a design decision here, not an omission, and it is recorded as one.

#### 3.1 Fish and Literature

The published material on this game is human and informal, and neither of its two substantial sources is quantitative or describes a program. McLeod’s Pagat entry [1] is the canonical rules source and records several points on which dialects differ; ours is specified in §2.2. Develin’s chapter [2] is the most detailed strategy writing we located, describes the same dialect, and supplies two observations this design uses: an ask teaches the opponents as much as it teaches a teammate, and a player is better off not declaring a half-suit it holds entirely, since holding one carries no risk. The second is a tactic and not a rule: the engine imposes no such condition, and §2.1 states what it does impose. Secondary strategy pages [3, 4, 5] restate blackballing, asking the asker, and an endgame delegation heuristic; they are evidence about how the game is played and talked about, not sources of quantitative claims.

**Computational prior art remains close to absent.** Searches across this project’s cycles found no academic work on this game, no public repository containing search, planning or a working learner for it, and no coverage in RLCard [9]. The one agent located, Somani’s *literature* [6, 7], plays the four-player, 48-card variant, which does not raise the six-player allocation problem, and publishes no strength numbers; a closed-source mobile application advertises bots with no stated methodology [8]. Every novelty statement in this paper therefore takes the form “we found no prior published work that ...” and should be read against that scope. One recent item is worth naming because a shared testbed is the kind of external calibration this line of work lacks: Valet packages twenty-one traditional imperfect-information card games in a common description language with measured branching factors and durations [10]. No comparison against it is attempted here.



### 3.2 Determinization, and why double-dummy analysis fails here

Determinization, or perfect-information Monte Carlo (sample hidden cards consistently with the public history, solve the resulting perfect-information game, and play the move with the best average value), is the standard recipe for hidden-hand card games [11, 12], and is unsound however many worlds are drawn. Frank and Basin named the two mechanisms: *strategy fusion*, in which the searcher plays a different strategy in each sampled world although a real player must commit to one strategy per information set, and *non-locality*, in which a node’s value depends on parts of the tree outside its own subtree [13, 14]. Long, Sturtevant, Buro and Furtak characterise when the method nevertheless performs well, through leaf correlation, bias and a disambiguation factor [15]; Information Set Monte Carlo Tree Search replaces independent trees with a single tree over information sets [16, 17]; the  $\alpha\mu$  family backs up Pareto fronts of world-vectors instead of scalar averages, repairing both mechanisms, and reports Bridge 3NT declarer play at 60.2% to 63.0% with twenty worlds [18]; and knowledge-based paranoia search reasons about the worst case over deduced-consistent worlds rather than the average, reporting over 1,000 points on the extended Seeger scale in Skat [19].

Fish breaks the method for a reason none of that literature addresses, and §2.3 states it: the perfect-information game is degenerate rather than merely fused, so  $\alpha\mu$ ’s repair of unsoundness does not rescue the construction.

What this corpus adds is that the binding problem is not the abstract failure but *the statistic*. One determinization’s value is the final half-suit differential, whose spread is an order of magnitude larger than the differences between the leading candidates, so an argmax over sampled means selects on residual noise, and does so more confidently the more candidates it is offered. That is the optimiser’s curse rather than strategy fusion, and §4.3 reports it measured at 35.69 pp by a control that forces the same code to return the blueprint’s own pick. The transferable claim from that result is the guard, not the search.

### 3.3 Search in imperfect-information and cooperative games

States in an imperfect-information game do not have well-defined values, so depth-limited search needs something at the leaf that represents the opponent’s ability to choose. Brown, Sandholm and Amos supply it: let the opponent select among  $k$  continuation strategies at the depth limit, each yielding its own leaf-value vector, and require the searcher to be good against that choice [24]. The soundness argument is stated for two-player zero-sum games, and Fish’s team-level game presents three uncorrelated opponent seats at the leaf, so nothing is inherited; it is adopted here as a robustification heuristic with a measurable consequence, and audited rather than assumed.

Two lines address what a search may prune. Zhang and Sandholm solve subgames over low-order knowledge rather than the full common-knowledge closure, and prove that the naive form can *increase* exploitability [25]. That warning applies directly to the live-ask gate this policy family ships, which is a knowledge-limited pruning rule, and it is why the side-channel and pathology gates of §7.8 were built before any search work rather than after it. Subgame solving in adversarial team games builds a gadget over the team belief DAG and runs in polynomial time given a best-response oracle *when the blueprint is sparse* [26]; a Fish blueprint is not sparse, so that escape hatch does not open here.

In cooperative games the constraint transfers verbatim. SPARTA improves a blueprint purely at test time and distinguishes single-agent search, which assumes every other agent strictly follows the blueprint, from multi-agent search, in which all agents run the *same* common-knowledge procedure [28]. The moment a second teammate also searches, the first teammate’s belief over its partner is

computed under a policy that partner is no longer playing, and the improvement guarantee is void. Fish satisfies the requirement cheaply, for the reason §2.3 gives; the qualifier about randomisation is what §4.3 turns into an engine switch.

The full-solver line [37, 35, 36] established that test-time re-solving over public belief states is principled rather than a hack. The algorithms are declined here, because each requires iterating or learning over a closure whose cardinality at deal time is  $45!/(9!)^5 \approx 1.9 \times 10^{28}$  deals from one seat. Update equivalence recasts search as replicating a last-iterate-convergent update locally, needing no public-information enumeration [27]; its improvement guarantee is for common-payoff games, which Fish is not. Learned Belief Search [29] is redundant here, because this game’s exact belief is already computed and oracle-validated. Online planning by policy fine-tuning [30] requires gradient updates at decision time and does not transfer to a CPU-only engine with no network. The depth-limited counter-strategy line [31, 32] carries one warning this study honours, robust-response methods do not compose with limited-look-ahead solving off the shelf, and the learned-model and gadget-refinement work [33, 34] is recorded for its meta-lesson rather than its machinery: when a search subproblem has many near-equivalent optima, which one is selected is not a tie-break. That is the same observation §4.3 makes about an argmax resolved by array order.

The result that a search of this kind can *increase* exploitability is Zhang and Sandholm’s [25]. Where the other work above carries a guarantee at all, its conditions and this game’s fit to them are recorded above, method by method. What none of them supports is reading a win rate as a bound on exploitability, which is why this paper measures the two separately. Local best response gives a cheap lower bound: a small greedy search looking one action ahead and assuming checks thereafter showed each of the 2016 competition entrants it evaluated to be more than 3,180 mBB/h exploitable [38], the competition’s winning submission was not among them, and the authors say why. Its two documented pitfalls are carried into §11.8, restricting where the response may begin deviating understates exploitability, and a negative score proves nothing, and every exploitability figure in this paper is a lower bound within its response class.

### 3.4 Policy-aware inference

Skat engines enumerate the deals consistent with the public record and reweight them by a model of what each player would have done, a supervised per-card location model [21], or reach probabilities taken from a policy [22], and a dedicated inference module was reported worth +217 tournament points per 36 games to the engine Kermit on its own [20]. The policy prior of §5.5 is the same idea in a much simpler form: a fitted soft reweighting of the rules posterior, with two parameters and no learned opponent model.

This is the only literature that has measured the same trade-off, and it is where §4.3’s central negative result belongs. What is new there is the margin and its direction: in this game the exact conditional posterior is computable exactly by a tractable factorised dynamic program and is cheap, so the usual excuse for approximating does not apply, and yet substituting the exact posterior into an otherwise identical policy *loses*, on two independent seed banks.

Two adjacent lines are named, one of them with a single member this study uses. Belief modelling in cooperative games, the Bayesian Action Decoder’s public-belief formulation [40] and its factorised-prescription successor [44], is aimed at games whose exact belief is intractable, which is the opposite of the situation here. And of the exact-counting machinery this study does not need in order to approximate, Ryser’s inclusion–exclusion formula [58], Glynn’s sign-vector formula [59], the Jerrum–Sinclair–Vigoda chain [60], matrix scaling [61] and sequential importance sampling [62], the block structure of §5 removes the necessity for all five and four go unused. Matrix scaling is the exception: the deployed path replaces the exact count join with a Sinkhorn iteration of exactly that kind, not

because the exact count is unaffordable but to buy the two policy-prior parameters (§5.5). No learned belief model appears anywhere in this corpus, and that is a stated non-goal rather than an oversight.

### 3.5 Team games and common information

The formal statement of three teammates sharing a payoff, holding different information and unable to communicate is the common-information approach [52], discussed in §2.3. The team-game equilibrium literature is cited to delimit scope: the target concept for a Fish team is a team-maxmin equilibrium with correlation [53], connected to extensive-form correlation by Farina and coauthors [54] and made computable through the team belief DAG and the two-player conversion [55, 56], with subsequent work on hidden-role settings [57]. None of it is computed here. The deeper reason a Fish team resists the DAG is that its meta-player fails A-loss recall, for exactly one reason: each seat privately observes nine cards its teammates do not.

The cooperative-play line is cited for its evaluation discipline rather than its algorithms. Hanabi was proposed as a benchmark precisely because self-play scores overstate coordination [39], and the ad-hoc-teamplay results that followed [41, 42, 43, 45, 48, 49, 51] are the reason the cross-play measurement of §11.6 exists at all: a convention that collapses when partners are swapped is brittleness, and it is invisible to self-play evaluation. Two neighbouring results are cited for what they show about porting rather than as ad-hoc teamplay: joint policy search is the bridge-bidding analogue, scoring simultaneous changes at several information sets [47], and the Hanabi construction that most clearly fails to port is the hat-guessing code [46], which works because each receiver sees every other receiver’s hand, and in Fish no player sees any other player’s hand. The same lesson from a different direction is Cicero’s use of a human-regularised policy [50].

### 3.6 Evaluation methodology

The evaluation framework of §7 draws on a literature usually treated as plumbing. Every cell in this paper is a paired duplicate rather than an independent sample, on the model of duplicate bridge and of the Annual Computer Poker Competition’s duplicate matches and common seeds [63], whose protocols are also the model for a fixed opponent panel scored identically for every arm. The estimator-side alternatives, DIVAT [64] and AIVAT [65], subtract a learned or hand-built value function from a single sample instead; neither is used here, and the pairing is what makes the design competitive with them without introducing a fitted object. Abstraction pathology [66] is the standing warning that a model can be improved on its own metric and made worse in play, which §9.4.2 reproduces exactly. Re-evaluation of agent populations [67] and the rating literature [68, 69, 70] inform why this paper reports a worst case and a minimax regret over a shared panel rather than a single rating number. Recent work on gold-standard evaluation practice [71] is consistent with the preregistration and sealing discipline of §7.6, which this project adopted independently.

## 4 Background: the evolution of FishBot

SESTINA v1.0 is the seventh configuration in a lineage, and several of its components and most of its evaluation machinery exist because of what earlier versions found. This section is deliberately short. Its purpose is to explain why SESTINA v1.0 looks the way it does, not to re-report earlier studies; the detailed corrections those studies require are collected in Appendix E.

#### 4.1 v0.2 to v0.4: engine, inference and policy foundations

The first versions established the domain and the inference. A count-conditioned, empirically tuned linear policy in a scripting-language simulator became a rules-exact C++ engine carrying an exact reference posterior, a faster deployed approximation, a twenty-feature ask score and a value-based declaration rule. The durable contribution of that period is the inference object of §5: an exactly computable observer-conditioned deal posterior, cheap enough to run in all six seats at once.

That period also established a habit of evaluation this paper spends considerable effort undoing. Cells were a few thousand games against a handful of opponents, ablations were unpaired, and conclusions were drawn from point estimates.

#### 4.2 v0.5: a failure mode removed, and honestly reported as worth nothing

v0.5 diagnosed a two-question deadlock in mirror play, in which two policies alternately ask each other for cards neither will surrender until the game ends in misdeclaration, and removed it. It then reported that removing it was worth almost nothing in win rate.

That report is the honest ancestor of this one, and it is where the project’s central difficulty first became visible. A release that removes a genuine defect and gains no measurable strength is either at the ceiling of its policy class or is measuring badly, and those two diagnoses call for opposite responses.

#### 4.3 v0.6: stronger play, and a measurement crisis

The v0.6 cycle set out to recover several points of headroom that the accumulated design register described. It found almost none of it, and discovered in the process that much of the measurement machinery was unsound. Four results from that cycle bear directly on SESTINA v1.0.

**Exchangeable ties are irreducible, so the tie-break is a coordination decision rather than a strength decision.** Measured in the v0.5 mirror, the policy that cycle set out to diagnose, 54.74% of *contested* ask decisions end with two or more candidates scoring bit-for-bit identically, and the winner is whichever the enumerator emitted first. 93.77% of those ties are two cards of one half-suit at one target, and the exact posterior *separates* them in 0.00% of cases, meaning it assigns them equal probability in every case. The same measurement on the v0.6 mirror gives 53.80%, and that is the figure §6.2 and §9.4.4 use, because SESTINA v1.0 modifies v0.6 rather than v0.5. Every rule that ranks tied candidates by that probability therefore realises the same hit rate, and so does enumeration order. The largest item on the inherited design register was not a defect but a property of the information set. What remains addressable is the dependence on enumeration order itself, which is a correctness problem, and §6.2 is how SESTINA v1.0 addresses it.

**The exact posterior is a worse predictor than the deployed approximation.** Scored as predictors on identical states in the v0.5 mirror, the ranking is:

| Posterior                                                             | Log loss (nats) | Argmax hit rate |
|-----------------------------------------------------------------------|-----------------|-----------------|
| Exact, uniform deal prior                                             | 1.42246         | 47.94%          |
| Approximation, certificate half of the prior deleted ( $\theta = 0$ ) | 1.39083         | 49.99%          |
| <b>v0.5’s deployed approximation</b>                                  | <b>1.38218</b>  | <b>51.49%</b>   |

The exact object is the worst row, and it stays worst with the prior deleted outright as well ( $\theta = \varphi = 0$ : 1.39339 nats and 49.14%), so most of the log-loss gap has no behavioural content at all, though most of the hit-rate gap does. The certificate half of the prior is the gap between the second and third rows; the prior as a whole is the gap from the  $\theta = \varphi = 0$  figures to the third. Separately from the ordering, exactness in this game also forecloses behaviour, because the block construction of §5 has no parameter through which an opponent’s tendencies could enter. Exactness and behaviour-awareness are not available in the same object here. This closed an experiment three studies had listed as outstanding, and it established a methodological rule the project has applied since, namely that exact Bayesian inference is an assumption to be tested rather than a property to be assumed. §9.4.1 is the second occasion on which that rule paid.

**Determinized search suffers an optimiser’s curse, and a paired guard removes it.** v0.6 built a search that samples full deals from the public posterior and plays each forward with all six players reconstructed at their own information sets, rather than made clairvoyant, which §2.3 explains would be degenerate. Chosen by an unguarded argmax over sampled means, that search is 35.69 pp *worse* than the blueprint it searches from, measured against a control that forces the same code to return the blueprint’s own choice at identical budget, seed and sample size. The cause is the statistic rather than the search: one determinization’s return is the final half-suit differential, whose spread is an order of magnitude larger than genuine differences between candidates, so the argmax selects on residual noise and does so more confidently the more candidates it is offered. A paired lower-confidence-bound deviation rule recovers all of it. The guarded search, which v0.6 measured and then deployed switched off, is a component of SESTINA v1.0 (§6.7).

**The evaluation could not resolve a one-point effect, and the optimiser could not move a policy.** Pooling every same-pair cell in the v0.6 battery gave 50.53% with a naive  $z$  of 1.13, a null that an earlier draft of that report published as one. Re-run at high power across 7 disjoint banks, the same comparison is 50.89% over 126,000 games [50.61, 51.16], above parity on all 7 banks. The effect had always been present; the instrument could not see it. In the same cycle, five mechanisms cleared a 95% paired interval at the per-cell budget the project had used since v0.3 and returned exactly zero at three times that budget. Separately, the fitting harness was found to be incapable of moving a policy at all, using one step size for thirty-four coordinates spanning two and a half orders of magnitude, an unpaired objective, and an objective documented as a worst case that was in fact a weighted mean.

#### 4.4 What v0.6 handed to the v0.7 programme

A policy roughly nine tenths of a point stronger than its predecessor; a design register most of whose entries had just been closed as worth nothing; a repaired optimiser; and an explicit diagnosis that at the budget the project had always used, the policy class looked flat without it being known whether the class *is* flat or the instrument was still too coarse.

The v0.7 programme was organised to answer that question in a way that could not be argued into a positive result. Five phases ran under separate briefs, each starting from a cleared context: characterise the instrument and establish what it can detect; search for adversaries in the open against the v0.6 frontier rather than against the deployed policy; build and test candidate mechanisms; select and freeze exactly one configuration; and evaluate that configuration on sealed material under a protocol registered in advance. §9 reports the first three phases and §11 the last.

**Table 1:** Notation used in §5 and §6.

| Symbol                                  | Meaning                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------|
| <i>The game and the hidden state</i>    |                                                                                            |
| $N = 6, M = 9,  H  = 6$                 | seats, half-suits, cards per half-suit; the deck is $M H  = 54$ cards                      |
| $s$                                     | the acting seat; $\mathcal{T}_s$ its two teammates, $\mathcal{O}_s$ its three opponents    |
| $D$                                     | the hidden state: the initial deal, an assignment of all 54 cards to seats, nine each      |
| $\mathcal{I}_s^t$                       | seat $s$ 's information set after $t$ public events                                        |
| $\mathcal{P}_t$                         | the running hash of the public event stream                                                |
| $\text{own}_t(c), H(c)$                 | the seat holding card $c$ ; the half-suit containing $c$                                   |
| $\mathcal{H}_t$                         | the half-suits not yet claimed                                                             |
| <i>Beliefs</i>                          |                                                                                            |
| $\mathcal{M}_t(c)$                      | the candidate set of card $c$ : seats not yet excluded by a certificate                    |
| $U_t, n_p$                              | the unresolved cards; the unresolved cards seat $p$ holds                                  |
| $B_h, \mathcal{F}_h$                    | the block of half-suit $h$ ; its assignments satisfying every certificate                  |
| $\sigma(\alpha), T_h(\mathbf{v})$       | an assignment's per-seat count signature; the block count table                            |
| $F_k, G_k$                              | the forward and backward partial convolutions of the count join                            |
| $Z, \mu(c, p)$                          | the number of consistent deals; the exact posterior marginal                               |
| $\hat{\mu}(c, p)$                       | the deployed approximation to $\mu(c, p)$                                                  |
| $\theta, \varphi$                       | the two policy-prior parameters of (7)                                                     |
| <i>Acting</i>                           |                                                                                            |
| $\mathcal{A}_t, a = (c, q)$             | the legal ask set; an ask naming card $c$ and opponent $q$                                 |
| $p_a$                                   | $\hat{\mu}(c, q)$ , the modelled hit probability of ask $a$                                |
| $\phi(a), w, u(a)$                      | the ask feature vector, its fitted weights, and the resulting score                        |
| $\lambda, \beta$                        | the linear and value scale parameters of (10)                                              |
| $M_H, q_H$                              | opponent-held mass and residual ambiguity in half-suit $H$ , in (17)                       |
| $\mathcal{E}_t, \varepsilon$            | the bit-for-bit tied candidate set and its tolerance                                       |
| $\pi_{\text{alloc}}, \pi_{\text{team}}$ | probability a named allocation is exactly correct; that the team owns a half-suit outright |
| $\delta, \tau$                          | the declaration value margin and the forced-endgame threshold                              |
| $\chi_t, \nu_t, K$                      | the deduction-state digest, the no-progress counter, its threshold                         |
| $\Phi_t$                                | the potential function of the termination argument in §6.6                                 |
| $D_{\text{det}}, w_d$                   | the number of sampled worlds and their importance weights                                  |
| $\kappa, \eta$                          | the search's deviation threshold and its blueprint blend, $\eta = 0$ here                  |

## 5 Exact observer-conditioned deal inference

Every agent in this lineage rests on an inference object that this section defines formally. The method is an exact enumeration carried out through a factorised dynamic program. It is not an analytic closed-form expression, and the paper avoids that phrase.

### 5.1 Notation

Table 1 fixes the symbols used here and in §6. A reader may prefer to skim it and return.



## 5.2 The engine, and the barrier between rules and policy

The engine is a single-header C++20 core. A hand is one 64-bit word over the 54 card indices  $c = 6h + i$  for half-suit  $h$  and offset  $i$ , so hand arithmetic is bit arithmetic. Random streams are counter-based and keyed by the triple (seed, seat, purpose), which gives the property everything else depends on: a deal and its entire play reproduce from the seed alone. The duplicate-block design of §7.2 rests on that property and on nothing else.

An acting policy never receives the game state. It receives its own knowledge object, its own hand and the public event stream. There is no reference from a policy to the deal, to another seat's hand or to the referee. This is an architectural barrier rather than a convention, and it is what allows the side-channel audit of §7.10 to be read as a test of the policy: a policy that cannot reach the deal cannot read it, so any decision the audit fails to reproduce from public information indicates state carried across deals rather than information taken from the referee.

## 5.3 What a seat knows, as constraints on the deal

Because a card changes hands only through an observed ask, the public record determines  $\text{own}_t(\cdot)$  given  $D$ . The only hidden object is therefore  $D$  itself. Write  $\mathcal{D}$  for the set of all deals; from one seat's viewpoint at deal time  $|\mathcal{D}| = 45!/(9!)^5 \approx 1.9 \times 10^{28}$ .

Each public event imposes a *hard* constraint, meaning a logical restriction on  $\mathcal{D}$  rather than a probabilistic tilt. The engine maintains them as a per-card candidate set  $\mathcal{M}_t(c)$ , initialised to all six seats and only ever shrunk:

**(C1) Own hand.** For each  $c$  in seat  $s$ 's hand,  $\mathcal{M}(c) \leftarrow \{s\}$ ; for every other card,  $s$  is removed from  $\mathcal{M}(c)$ .

**(C2) Card counts.** Every seat's current hand size is public, so with the transfer record each seat's count of unresolved cards,  $n_p$ , is known. This is the only constraint that couples different half-suits.

**(C3) Transfers.** A successful ask makes  $\text{own}(c)$  public from then on, so  $\mathcal{M}(c)$  collapses to a singleton.

**(C4) Failed asks.** If  $q$  was asked for  $c$  at time  $r$  and did not hold it, then  $q \notin \mathcal{M}(c)$  unless  $q$  has acquired  $c$  since, which is public.

**(C5) Ask legality.** An asker of  $c$  held another card of  $H(c)$  at that moment and did not hold  $c$ . The first is a disjunction over the other five cards of the half-suit and is the one constraint that is not a simple exclusion.

Conditioned on  $\mathcal{I}_s^t$ , all consistent deals are equally likely, so the posterior is uniform on the feasible set:

$$\Pr[D \mid \mathcal{I}_s^t] = \frac{\mathbb{1}[D \models \mathcal{I}_s^t]}{Z}, \quad Z = |\{D \in \mathcal{D} : D \models \mathcal{I}_s^t\}|. \quad (1)$$

The modelling content is entirely in which constraints are imposed. Nothing here refers to how anyone plays, which is the property §5.5 then gives up deliberately.

## 5.4 Factorisation by half-suit, joined by a count dynamic program

Computing  $Z$  by enumerating  $\mathcal{D}$  is infeasible, and a constraint system of this shape generally induces a permanent-like count, the object the literature approximates by inclusion–exclusion [58], sign-vector formulae [59], Markov chains [60], matrix scaling [61] or sequential importance sampling [62].

Fish avoids that, because every constraint except (C2) is confined to a single half-suit. Constraints (C3), (C4) and (C5) all speak about one card or one half-suit; only the per-seat capacities  $n_p$  couple half-suits, and they couple them only through counts.

Partition the unresolved cards by half-suit,  $U_t = \bigcup_h B_h$ , where  $B_h$  is the *block* of half-suit  $h$ . For a block, let an *assignment* be a map  $\alpha : B_h \rightarrow \{0, \dots, 5\}$  with  $\alpha(c) \in \mathcal{M}(c)$ , and let  $\mathcal{F}_h$  be the assignments that additionally satisfy every (C5) disjunction touching  $h$ . Write the *count signature* of an assignment as the vector

$$\sigma(\alpha) = (|\alpha^{-1}(0)|, \dots, |\alpha^{-1}(5)|) \in \mathbb{Z}_{\geq 0}^6.$$

**Stage 1: block enumeration.** For each half-suit independently, enumerate  $\mathcal{F}_h$  and accumulate a table indexed by count signature,

$$T_h(\mathbf{v}) = |\{\alpha \in \mathcal{F}_h : \sigma(\alpha) = \mathbf{v}\}|, \quad T_h^{c \rightarrow p}(\mathbf{v}) = |\{\alpha \in \mathcal{F}_h : \sigma(\alpha) = \mathbf{v}, \alpha(c) = p\}|. \quad (2)$$

A half-suit holds six cards and resolved ones are removed, so  $|B_h| \leq 6$  and the enumeration is small.

**Stage 2: the count join.** Constraint (C2) says the signatures must sum to the vector of per-seat capacities  $\mathbf{n} = (n_0, \dots, n_5)$ . Order the blocks arbitrarily and define the partial convolution. A half-suit carrying no live (C5) disjunction factorises further into one independent unit per card, so the implementation runs one layer per unit rather than one per half-suit; the recursion below is the same object either way.

$$F_0(\mathbf{0}) = 1, \quad F_k(\mathbf{v}) = \sum_{\mathbf{u}} F_{k-1}(\mathbf{v} - \mathbf{u}) T_{h_k}(\mathbf{u}), \quad (3)$$

so that

$$Z = F_M(\mathbf{n}). \quad (4)$$

Running the same recursion backwards gives suffix tables  $G_k$ , and the marginals follow by substituting one block's card-resolved table for its plain one:

$$\mu(c, p) = \frac{1}{Z} \sum_{\mathbf{u}, \mathbf{v}} F_{k-1}(\mathbf{v}) T_{h_k}^{c \rightarrow p}(\mathbf{u}) G_{k+1}(\mathbf{n} - \mathbf{v} - \mathbf{u}), \quad c \in B_{h_k}. \quad (5)$$

Three quantities the policy needs are *defined* by this machinery, and §5.5 records that the deployed configuration substitutes a cheaper approximation for each of them. The per-card marginals are (5). The probability that team  $\mathcal{T}$  owns half-suit  $h$  outright,  $\pi_{\text{team}}(h)$ , is (5) restricted to block assignments whose image lies in  $\mathcal{T}_s \cup \{s\}$ . And the probability of a fully named allocation, which is exactly what a declaration asserts, is

$$\pi_{\text{alloc}}(h, \alpha^*) = \frac{\mathbb{1}[\alpha^* \in \mathcal{F}_h]}{Z} \sum_{\mathbf{u}, \mathbf{v}} F_{k-1}(\mathbf{v}) \mathbb{1}[\sigma(\alpha^*) = \mathbf{u}] G_{k+1}(\mathbf{n} - \mathbf{v} - \mathbf{u}), \quad (6)$$

for the single assignment  $\alpha^*$  the declaration names. The leading indicator is not cosmetic: an assignment with a surviving count signature need not itself satisfy the half-suit's certificates, so feasibility is checked explicitly. No sampling is involved in any of the three.

**Cost.** The signature space is bounded by the capacities, each block table has at most  $6^{|B_h|}$  entries before pruning, and the join is  $M$  convolutions over that space. An exact rebuild costs 0.4 ms, which is why this project has been able to treat inference as solved and spend its cycles on policy.

## 5.5 The deployed approximation, and the policy prior it adds

Equation (1) is exact *under a uniform prior over constraint-consistent deals*. It contains no model of behaviour, because the construction above consumes the deduction state alone and has no parameter through which an opponent’s tendencies could enter.

The path the agent deploys adds two such parameters and gives up exactness for them. It replaces the count join by iterative proportional fitting, better known here as Sinkhorn scaling, on the  $|U_t| \times 6$  matrix of admissible card–seat pairs. Initialise

$$\hat{\mu}^{(0)}(c, p) = \mathbb{1}[p \in \mathcal{M}(c)] \cdot \exp\left(\text{clip}(\theta a_{p,H(c)} - \varphi (A_p - a_{p,H(c)}), \pm 2.6)\right), \quad (7)$$

where  $a_{p,h}$  is the number of asks seat  $p$  has made in half-suit  $h$  and  $A_p$  its total asks, and then alternate the two normalisations

$$\hat{\mu}(c, \cdot) \leftarrow \frac{\hat{\mu}(c, \cdot)}{\sum_p \hat{\mu}(c, p)}, \quad \hat{\mu}(\cdot, p) \leftarrow \hat{\mu}(\cdot, p) \cdot \frac{n_p}{\sum_c \hat{\mu}(c, p)}, \quad (8)$$

to a fixed iteration count. The row step makes each card’s distribution a probability; the column step enforces the capacity constraint (C2) that the exact join enforces combinatorially.

Between outer sweeps the routine also applies an independence conditioning step for each live (C5) disjunction: for a certificate saying that seat  $j$  held *some* card of a half-suit, each candidate card  $c'$  of that half-suit is rescaled to  $\hat{\mu}(c', j) / (1 - \prod_{c''} (1 - \hat{\mu}(c'', j)))$ , the product running over the half-suit’s cards  $c''$ . So (C5) is neither enforced exactly nor discarded in the deployed path: it is imposed approximately, under an independence assumption the exact enumeration of §5.4 does not need. The deployed parameters are  $\theta = 0.37062$  and  $\varphi = 0.14525$ .

Equation (7) is the entire behavioural content of the agent’s beliefs: a seat that has asked repeatedly in a half-suit is treated as more likely to hold cards of it, because asking requires already holding one. It is a soft version of the (C5) certificate, in the spirit of the Skat inference line [21, 22, 20] but far simpler, with two parameters and no learned opponent model.

§4.3 reports the consequence, which is the most counterintuitive result in the lineage: scored as predictors on identical states,  $\hat{\mu}$  beats  $\mu$ . Exactness in this game is purchased by discarding (7), and (7) is worth more than exactness. That is why the paper treats “exact Bayesian inference” as an assumption to test rather than a property to assume, and §9.4.1 is the second occasion on which the rule paid.

## 5.6 Verification

Three checks run over the engine, and all three ran as preconditions of the evaluation in §11.

**Soundness audit.** Every seat’s candidate set is checked against the true deal at every decision:  $\text{own}_t(c) \in \mathcal{M}_t(c)$  must hold always, since no agent’s deduction may exclude the truth. The verification run for this evaluation recorded 0 violations in 6,737,436 checks, and the preceding cycle 0 in 23,594,580.

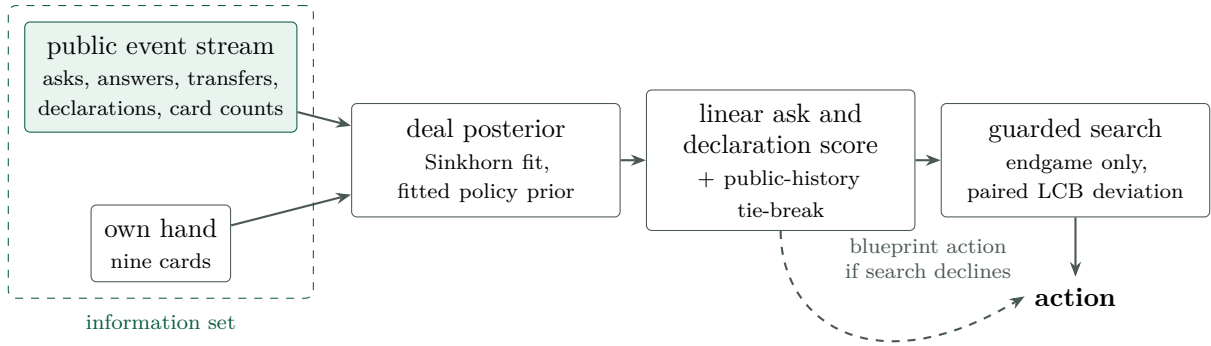
**Cross-engine agreement.** The block dynamic program, an independent card-counting path and an exact sampler must agree on (5). The self-test reports 424,986 checks passing, with block-versus-card agreement at the limit of double precision.

**Identity control.** The current policy, carrying its predecessor’s parameter vector with every new mechanism disabled, must produce a transcript identical to its predecessor’s under md5. It does (PASS). Without this an ablation is not an ablation, because a measured difference could be an artefact of refactoring.

Two defects these checks found are recorded here because both changed previously published numbers, and both survived for the same reason: they were invisible to the path the agent actually deploys. The block dynamic program parked every instance’s tables in a shared per-thread pool, so a second construction on the same thread silently repointed the first instance’s tables, giving 285 mismatches in 294 checks; any earlier number computed with the exact belief in a configuration where more than one seat held such an object is affected, and the deployed approximate-belief numbers are not. Separately, an audit that verifies the declaration pre-gate never rejects a correct declaration had been parsed only in an older branch and so had never run; once enabled it reports 0 false negatives over 5,472,906 declaration opportunities. Appendix E carries the full register.

## 6 SESTINA v1.0

SESTINA v1.0 is a configuration of the v0.6 policy family rather than a new architecture. That sentence was written into the evaluation protocol before any holdout result was known, and the evaluation did not disturb it. This section describes the agent conceptually: Figure 1 shows the decision pipeline, and Table 2 sets its components against v0.6’s. The literal configuration string, its option map, the parameter vector layout and the freeze artifact’s hashes are in Appendix B.



**Figure 1:** The information structure of a Fish seat and the SESTINA v1.0 decision pipeline. A seat observes the public event stream and its own hand, and nothing else; there is no path from a policy to the deal or to another seat’s hand (§5.2). Because every card movement is public, that information set determines an exact posterior over the initial deal (§5). The deployed path does not construct it: it runs a cheaper approximate fit that starts from a policy prior and imposes the same constraints iteratively (§5.5). The search runs only in the endgame and departs from the blueprint’s choice only when a paired lower-confidence bound justifies it (§6.7).

### 6.1 The decision problem, and the ask policy

At its turn a seat faces one decision with two branches: declare a half-suit, or ask. Declaration is treated first in the code and is described in §6.4; this subsection covers the ask.

**The legal action set.** With  $\mathcal{O}_s$  the opponents and  $\mathcal{H}_t$  the half-suits not yet claimed, the rules of §2.1 give

$$\mathcal{A}_t = \{(c, q) : q \in \mathcal{O}_s, \text{hand}_t(q) \neq \emptyset, H(c) \in \mathcal{H}_t, c \notin \text{hand}_t(s), \text{hand}_t(s) \cap H(c) \neq \emptyset\}. \quad (9)$$

The last condition is what makes every ask informative: the asker must already hold a card of the half-suit, so asking publishes partial ownership.

**Table 2:** Components of SESTINA v1.0 against v0.6. Harness identifiers appear in parentheses and are used in the paper only where a command must be reproduced.

| Component                             | v0.6                                                           | SESTINA v1.0                                                                                        | Section |
|---------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------|
| Deal inference                        | Sinkhorn fit to the deal posterior, from a fitted policy prior | unchanged                                                                                           | §5      |
| Ask and declaration policy            | fitted linear score, 37 coordinates                            | unchanged; no refit in this cycle                                                                   | §6.1    |
| Tie-breaking among equal-scoring asks | enumeration order (unstable sort)                              | hash of the public event stream ( <code>rtie</code> )                                               | §6.2    |
| Half-suit contestation                | absent                                                         | contested-half-suit weight ( <code>r12</code> )                                                     | §6.3    |
| Declaration urgency escalation        | event-count clock at 220 events                                | disabled ( <code>pool</code> , <code>oppfloor</code> , <code>force</code> , <code>askfloor</code> ) | §6.5    |
| Termination safeguard                 | the urgency clock                                              | deduction-state stall detector ( <code>stall</code> )                                               | §6.6    |
| Test-time search                      | built and measured, deployed disabled                          | enabled, endgame-truncated, paired LCB guard                                                        | §6.7    |

What the policy actually enumerates is a strict subset of (9). A *live-ask filter* first removes candidates the seat can already prove are misses, that is, those whose target is excluded by a certificate,  $q \notin \mathcal{M}(c)$ . It is this filter, rather than any weighting inside (10), that keeps a provably dead ask off the score. It binds: it removes what would otherwise have been the argmax at roughly 1.4% of 269,423 ask decisions captured from SESTINA v1.0’s own seats. If no candidate survives the filter the full legal set is restored, so the guarantee is not absolute.

**The score.** Each candidate is scored by a fitted linear function of a 20-dimensional feature vector  $\phi(a)$ , plus a learned value term and, in SESTINA v1.0, three additional ownership terms:

$$u(a) = \lambda w^\top \phi(a) + \beta V_{\text{ask}}(a) + \lambda(w_{\text{void}} g_1(a) + w_{\text{team}} g_2(a) + w_{\text{last}} g_3(a)), \quad (10)$$

with  $\lambda = 0.78874$  and  $\beta$  scale parameters of the fit. SESTINA v1.0 adds a fourth block of 18 extended coordinates, of which 1 is non-zero; that one is (17) and §6.3 gives it. Writing  $p_a = \hat{\mu}(c, q)$  for the modelled hit probability, the features  $\phi$  are functions of the belief and the public state. They include  $p_a$  and  $p_a^2$ ; an indicator that the card is publicly certain; the acting seat’s holding in  $H(c)$  and the posterior mass its team already carries; the best continuation probability elsewhere in the half-suit; the reply threat and exposure a miss would create, both weighted by  $1 - p_a$ ; the target’s hand size and an indicator that this ask would empty it; the binary entropy  $H_2(p_a)$ ; and a score-margin term. Appendix B lists all twenty with their fitted weights.

The three extra terms are the ones v0.6 added, and each expresses an effect the twenty could not:

$$g_1(a) = p_a \prod_{c' \in H(c) \setminus \{c\}} (1 - \hat{\mu}(c', q)), \quad (11)$$

$$g_2(a) = \Pr[\text{own}(c) \in \mathcal{T}_s \cup \{s\}], \quad (12)$$

$$g_3(a) = p_a \cdot \mathbb{1}[q \text{ is the last opponent holding cards, and holds one}]. \quad (13)$$

Equation (11) is the probability that the ask takes the target’s *last* card of the half-suit, (12) the posterior mass the acting team already carries, so a measure of how wasteful the ask is, and (13) an

indicator that the ask would trigger the forced endgame. Equation (11) prices information denial in its bluntest form. Because a legal ask requires holding a card of the half-suit, removing a target’s last card of  $H(c)$  permanently removes that target’s right to ask in it.

**Selection.** The policy takes the argmax, with the tie handling of §6.2 and, when enabled, the search of §6.7:

$$a^* = \arg \max_{a \in \mathcal{A}_t} u(a). \quad (14)$$

SESTINA v1.0 inherits  $w$ ,  $\lambda$ ,  $\beta$  and the three extra weights from v0.6 unchanged. **No gradient or population fit was run in this cycle:** the one coordinate that is not inherited, (17), had its value chosen as the argmax of a measured dose response, which is a one-dimensional selection on win rate and is reported as such in §11.2.3. The vector was found by v0.6’s repaired optimiser under a minimax-regret objective over a four-member opponent panel. Of the 55 coordinates in the extended vector a number are known to be inert, and Appendix B lists which and why.

## 6.2 Tie-breaking that preserves common knowledge

In the v0.6 mirror, 53.80% of contested ask decisions end with two or more candidates scoring bit-for-bit identically under (10) (§4.3), and the winner was whichever the enumerator emitted first, under an unstable sort. That is a correctness problem rather than a strength problem. The choice is arbitrary, but it is also unpredictable, and §2.3 explains why unpredictability of the wrong kind is expensive here: three identical deterministic teammates coordinate for free only if each can compute what the others will do, and a tie broken by an implementation detail is not something a teammate can reconstruct.

Order the candidates by score and let the tied set be

$$\mathcal{E}_t = \{a \in \mathcal{A}_t : u(a) \geq u(a_{(1)}) - \varepsilon\}, \quad (15)$$

where  $a_{(1)}$  is the top-ranked candidate and  $\varepsilon$  is a fixed tolerance. SESTINA v1.0 replaces enumeration order with a deterministic draw keyed by the public record:

$$a^* = \mathcal{E}_t[\iota(\rho, |\mathcal{E}_t|)], \quad \rho = \text{hash}(\mathcal{P}_t, s, t), \quad (16)$$

with  $\mathcal{P}_t$  the running hash of the public event stream,  $s$  the seat index,  $t$  the event count, and  $\iota$  an index map from the hash into the tied set. Every seat observes all three arguments, so every teammate can recompute  $\rho$ ; no seat’s private hand enters it, and the choice is reproducible from the public record rather than being an artefact of the enumerator’s sort order.

Two scope conditions belong with that. (16) governs the non-search path only: when the endgame search fires it supplies its own selection (§6.7) and index 0 is whatever the sort left first. And the search draws its determinizations from a private per-seat stream seeded from the harness, so the policy is a deterministic function of (own hand, public record, reset seed) rather than of the first two alone. That third argument is exactly what the side-channel audit of §7.10 replays, which is why the audit is a test of seat isolation and not of determinism.

The engine also offers a variant that draws  $\rho$  from a private per-seat stream. That variant is *not* used: it would break the reconstructibility that makes (16) a coordination device. §9.4.4 reports what happened when a later experiment tried to *select* within  $\mathcal{E}_t$  rather than randomise over it.



### 6.3 Half-suit contestation

The one component carrying measurable strength is a single additional coordinate, one non-zero entry out of 18 in an extended block of the parameter vector. Let  $M_H$  be the expected number of cards of half-suit  $H$  held by the opposing team, and  $q_H$  the number whose holder this seat cannot place. The term is

$$r(a) = (1 - p_a) \cdot \frac{M_{H(c)}}{6} \cdot \frac{q_{H(c)}}{6}, \quad (17)$$

entering (10) with a positive coefficient of 25, so a large value *encourages* the ask. It selects asks that are likely to miss, into half-suits the opponents dominate, whose ownership is still ambiguous.

**The mechanism is not the one the coordinate was written for, and saying so matters.** The feature was designed to price *information denial*: a failed ask publishes a negative certificate, and the intent was that the fitted coefficient would come out negative, meaning refuse to hand that certificate over. It came out positive. The negative branch was swept separately and is flat to negative throughout, reaching  $-1.47$  pp at the mirror-image dose, so the design intent is **refuted rather than merely unexplored**.

What the positive branch does was measured on the target rather than inferred. Over  $\sim 261,000$  target ask decisions, the target’s ask accuracy falls from 0.540 to 0.521, while its declaration accuracy *rises* and it becomes *less* often urgent, which rules out the declaration channel where this phase expected to find everything. The reading the artifact supports is contestation: asking into half-suits the opponents are trying to complete degrades the quality of their asks, and the acting team pays part of the same cost, its own ask accuracy falling from 0.543 to 0.534. Three sibling coordinates written to express the same denial idea were measured at the same dose and are inert or harmful, which is what pins the effect to this combination rather than to opponent mass in general.

Two facts about its provenance matter for §11.4. It was found by a coordinate sweep during adversary generation rather than by architecture work, and it is hand-set: the value 25 is the argmax of a dose response measured at 48,000 games a dose, not the output of a gradient or population fit. And it is not uniformly good. Against the then-deployed policy it scored  $+2.71$  pp, but its worst cell over the v0.6 frontier is  $-1.69$  pp against **Fsearch**, which is negative, so it is a strength gain with a worst case rather than an exploit (§9.3).

### 6.4 The declaration rule

A declaration names, for each of the six cards of a half-suit, which member of the declaring team holds it. It is all-or-nothing: correct, and the team scores the half-suit; wrong, and the opponents do. The decision therefore has two parts, choosing what to claim and choosing when.

**What to claim.** For half-suit  $h$  the agent takes the most probable allocation among its own team,

$$\alpha^*(h) = \arg \max_{\alpha: H \rightarrow \mathcal{T}_s \cup \{s\}} \pi_{\text{alloc}}(h, \alpha), \quad (18)$$

with  $\pi_{\text{alloc}}$  from (6). (18) is the object; it is not what the deployed path computes. SESTINA v1.0 enumerates the capacity-feasible assignments of the unresolved cards of  $h$  to its own team and takes the argmax of the *product of per-card marginals*,  $\prod_c \hat{\mu}(c, \alpha(c))$ . That product is the selection score only. The winner is then priced separately, by sequential conditioning: the belief is refit after each card of the allocation is fixed in turn, and the resulting chain-rule product is what the policy reports as  $\pi_{\text{alloc}}^*$  and thresholds on. The exact object (6) is available from the block dynamic program but

sits on no decision path. §9.4.1 reports the measurement that closed the question of whether that shortcut costs anything: over ~210,000 recorded declarations it selects the same allocation as the *sequential joint*’s argmax every time, because that joint’s first chain-rule factor *is* the marginal and the renormalisation that follows reweights the survivors without reordering them.

**When to claim.** Write  $\pi_{\text{alloc}}^* = \pi_{\text{alloc}}(h, \alpha^*)$  and  $\pi_{\text{team}}$  for the probability the team owns  $h$  outright. The value function is written on the assumption that waiting is free, on the ground that an opponent cannot claim a half-suit the acting team provably holds. That ground binds strictly only in the locked case, and the rule is applied down to a team-ownership probability well below one, so the assumption is an approximation rather than an identity. Under it the decision is not a race but a trade: declaring scores a point now with probability  $\pi_{\text{alloc}}^*$ , and publishes the team’s holdings either way. The agent evaluates that with its learned state-value function  $V$ :

$$\Delta = \underbrace{\pi_{\text{alloc}}^* V(\text{state} \mid \text{correct}) + (1 - \pi_{\text{alloc}}^*) V(\text{state} \mid \text{wrong})}_{\text{declare}} - \underbrace{V(\text{state})}_{\text{wait}}, \quad (19)$$

and declares when  $\Delta > \delta$  for a fitted margin  $\delta$ . Both conditioned states differ from the waiting state in having the *unresolved* cards of  $h$  removed from the unresolved set, typically fewer than six, and in the deltas of the four half-suit-control features the value function reads, all of which are functions of the team’s expected share of  $h$  before the claim. They differ from *each other* only in the score margin, by +1 and −1.

Two overrides sit above (19). Under the forced endgame, when the opposing team is cardless and the acting team must resolve the remaining half-suits among itself, the agent declares on a threshold,  $\pi_{\text{alloc}}^* \geq \tau$ , and on a lower threshold once  $\pi_{\text{team}} > 1 - 5 \times 10^{-4}$ , since a half-suit the team certainly owns will have to be claimed eventually and delay only costs information. Under *pressure*, the escalation described next, stage 1 lowers the thresholds and stage 2 removes them: the best available allocation is declared unconditionally, whatever its probability. SESTINA v1.0 removes the mechanism that used to raise pressure in ordinary play, and §6.6 is what now supplies stage 2.

## 6.5 Removing the urgency escalation

v0.6 inherited a declaration-urgency escalation keyed to an event counter. Past a threshold of 220 public events the declaration rule lowers its team-ownership floor and begins cashing half-suits at close to a coin flip, which costs 15.18 pp to a policy held in that regime. No adversary reaches it, since the longest game any adversarial search produced against an unmodified opponent is 149 events, but the machinery is defective in ordinary play as well: §9.4.1 reports that removing it eliminates 71% of the entire declaration allocation-error class. SESTINA v1.0 disables it (`pool`, `oppfloor`, `force`, `askfloor`).

The escalation was, however, also the policy’s termination safeguard, so removing it creates an obligation.

## 6.6 A stall detector as the replacement termination safeguard

The escalation SESTINA v1.0 removes was keyed to a clock. The condition the clock stood in for is “nothing is happening any more”, and that condition is directly observable from the seat’s own deduction state, which is both tighter and safer.

Let  $\chi_t = \text{hash}(\text{own}_t(\cdot), \mathcal{M}_t(\cdot), U_t, (\text{hand sizes})_t, (\text{live half-suits})_t)$ , the digest of the hand certifies the seat holds. It deliberately *excludes* the list of outstanding (C5) disjunctions: a repeated

ask appends a duplicate disjunction carrying no information, so counting that list as progress would blind the detector to the one loop it exists to break. Define the no-progress counter

$$\nu_t = \begin{cases} 0, & \chi_t \neq \chi_{t-1}, \\ \nu_{t-1} + 1, & \chi_t = \chi_{t-1}, \end{cases} \quad (20)$$

and escalate to pressure stage 1 when  $\nu_t \geq K$  and to stage 2 when  $\nu_t \geq 2K$ , with  $K = 12$  and  $2K = 24$  in SESTINA v1.0.

**Why it is a safeguard and not a guarantee.** Consider  $\Phi_t = \sum_{c \in U_t} (|\mathcal{M}_t(c)| - 1) + |U_t| + |\mathcal{H}_t|$ . Certificates only accumulate: a seat excluded from a card’s candidate set is never restored, a resolved card is never unresolved again, and a claimed half-suit never returns. Every event that changes  $\chi$  by one of those routes therefore strictly decreases  $\Phi_t$ , which is a non-negative integer, so there can be only finitely many of them. **This argument does not close.** The digest also contains hand sizes, so a successful ask always counts as progress, and a card moving between two seats is not a monotone gain; nothing above forbids a card being won back later. The rule therefore bounds the mode of non-termination that actually occurs, a run of asks producing no new certificate, and not every mode. That is why this paper calls it a safeguard, and why §12.6 records termination as empirical.

The threshold was set from measurement. Over 800 mirror games v0.6 never produces a no-progress run longer than 6, while the deliberately deadlocked negative control of §7.8 produces 326. Any  $K \in [12, 60]$  is therefore simultaneously unreachable in ordinary play and immediate in a deadlock. Armed, the rule’s mirror transcript digest is byte identical to the same configuration without it: its cost in ordinary play is not small but identically zero, because  $\nu_t$  never reaches  $K$ . §11.4 confirms that on holdout material. Note also that  $\nu_t$  is a function of the seat’s own hand and the public record alone, so the rule opens no information channel.

## 6.7 Guarded determinized test-time search

SESTINA v1.0 enables the search v0.6 built, measured, and then deployed switched off. It runs only in the endgame, when at most 26 cards remain unresolved, and it revises the blueprint’s choice only when a paired statistical test says it should.

**Sampling worlds.** The search draws  $D_{\text{det}} = 12$  determinizations from the exact posterior of (1), using a per-card capacity dynamic program as a uniform sampler over deals satisfying (C1)–(C4), with the (C5) disjunctions enforced by rejection, so an accepted draw is an exact sample. This sampler is a distinct object from the block program of §5.4, which enumerates (C5) inside its blocks instead.  $D_{\text{det}}$  is a target rather than a guarantee: draws are attempted up to a cap and the rule proceeds with however many survive rejection. The policy prior of (7) is then applied as an importance weight rather than baked into the sampler, which keeps the sampler exact and the prior auditable:

$$\tilde{w}_d = \prod_{c \in U_t} \exp\left(\text{clip}(\theta a_{o_d(c), H(c)} - \varphi(A_{o_d(c)} - a_{o_d(c), H(c)}), \pm 2.6)\right), \quad w_d = \frac{\tilde{w}_d}{\sum_{d'} \tilde{w}_{d'}}, \quad (21)$$

where  $o_d(c)$  is the seat holding card  $c$  in world  $d$ .

**Rolling out.** The candidate set is the top 4 blueprint candidates, with index 0 reserved for the blueprint’s own choice  $a_{(1)}$ . For each candidate  $r$  and world  $d$ , all six seats are reconstructed *at their own information sets* under  $d$ , each playing a cheaper blueprint than the acting agent’s own: the v0.5 policy with an independent-marginal belief and no re-scoring pass and the game is played to a terminal state or until 12 further public events have elapsed. A run reaching the end returns the acting team’s half-suit differential; a run cut at the limit returns a *material* evaluation of the position instead, being the scored half-suit differential plus the team’s expected control of the live half-suits. That is v0.6’s own hand-written leaf, not a learned one: §9.4.2 reports what happened when a fitted replacement was built. Write  $V_{r,d}$  for either. This is the point at which the method departs from standard determinization: the continuation players are not made clairvoyant. §2.3 explains why that matters here, namely that clairvoyant play in Fish is degenerate and every action would score alike. All candidates share the same  $D_{\text{det}}$  worlds, so the comparison is paired and the noise that matters is the noise in the difference.

**The guarded deviation rule.** For each candidate  $r \geq 1$  form the paired difference against the blueprint,

$$m_r = \sum_d w_d (V_{r,d} - V_{0,d}), \quad \text{var}_r = \sum_d w_d (V_{r,d} - V_{0,d} - m_r)^2, \quad (22)$$

and with the Kish effective sample size  $n_{\text{eff}} = 1 / \sum_d w_d^2$  take

$$\text{se}_r = \sqrt{\frac{\text{var}_r}{n_{\text{eff}} - 1}}, \quad \text{LCB}_r = m_r - \kappa_r \text{se}_r + \eta (u(a_{(r+1)}) - u(a_{(1)})). \quad (23)$$

The search deviates to  $\arg \max_r \text{LCB}_r$  if that maximum is positive, and otherwise plays the blueprint’s choice. The shrinkage is  $\kappa = 2.5$ . The term  $\eta$  blends the blueprint’s own ranking back in and is **zero in this configuration**, so (23) reduces to a pure paired lower confidence bound.

Equation (23) is the whole of what makes the machinery usable. Chosen instead by an unguarded  $\arg \max_r m_r$ , the same search is 35.69 pp *worse* than the blueprint it searches from (§4.3): one world’s return is a final half-suit differential whose spread is an order of magnitude larger than genuine differences between candidates, so the maximum selects on residual noise, and does so more confidently the more candidates it is offered. Requiring  $\kappa$  standard errors of a *paired* difference removes that.

One detail is worth stating because §9.4.4 turns on it. Within the tied set  $\mathcal{E}_t$  of (15) the rule applies  $\kappa_r = 0$ , that is, no shrinkage at all, which is precisely where its selection signal is most noise-dominated.

## 6.8 How three teammates coordinate without communicating

SESTINA v1.0 deploys as three identical agents. They hold different private hands, they may not signal, and the only channels the dialect leaves are the forced-endgame willingness bit and the successor a cardless turn-holder names (§2.2), both of which §2.2 measures as empty or near-empty. So all coordination has to come from somewhere else.

It comes from determinism plus a shared observation. Let  $\Pi$  denote the policy, a map from (own hand, public record) to an action. Every teammate runs the same  $\Pi$  and observes the same public record  $\mathcal{P}_t$ . Teammate  $j$  cannot compute  $\Pi$ ’s output at seat  $i$ , because it does not know  $i$ ’s hand. But it knows  $\Pi$  itself, and it knows  $\mathcal{P}_t$ , so it can compute the *distribution* of  $i$ ’s action induced by its own posterior over  $i$ ’s hand,

$$\Pr[a_i = a \mid \mathcal{I}_j^t] = \sum_{D \models \mathcal{I}_j^t} \Pr[D \mid \mathcal{I}_j^t] \mathbf{1}[\Pi(\text{hand}_i(D), \mathcal{P}_t) = a], \quad (24)$$

and, having observed  $a_i$ , condition on it. That is exactly the (C5) reasoning of §5.3 carried one step further: a legality certificate says the asker held *some* card of the half-suit, while (24) says the asker held a hand on which  $\Pi$  preferred that ask. The deployed approximation does not evaluate (24) in full, which would require enumerating hands; it captures the first-order part of it through the ask-tally prior (7). The white-box adversary of §11.8.1 is the construction that evaluates (24) against SESTINA v1.0 rather than for it, and §9.2 reports what that inversion is worth.

Two conditions make this work, and both are engineering constraints rather than assumptions.  $\Pi$  must be *reconstructible*: a teammate reasoning about seat  $i$  must be able to evaluate  $\Pi(\cdot, \mathcal{P}_t)$  on a hypothesised hand. An unstable sort does not break determinism, since it is deterministic for a fixed input and build, but it does break reconstructibility, because the result depends on an implementation detail no teammate can model. (16) restores it on the non-search path. And  $\Pi$  must be common knowledge, which for three copies of one published agent it is. No shared secret and no pre-play randomisation are needed or used, which is the sense in which the team’s correlation device is public (§8).

**One defect the evaluation runs under.** Agents are constructed once per worker thread and reused across the deals that thread is given, and some per-agent state survives the inter-deal reset, reachable only under the search. This violates the premise that  $\Pi$  depends on (own hand,  $\mathcal{P}_t$ ) alone: a decision can depend on which deals the thread happened to process earlier. The carrier was identified as the rollout engine, which the reset never touches; the specific field was not. It was deliberately not repaired, because the one-line fix changes the play of every searching configuration by roughly one deal in 800, and applying it after the configuration was frozen would defeat the purpose of freezing. §11.9 measures its magnitude on holdout material, and §12.6 states what it costs the argument above.

## 6.9 The complete decision procedure

Collecting the pieces, one turn of SESTINA v1.0 at seat  $s$  and time  $t$  runs as follows.

1. **Update beliefs.** Fold the new public events into the candidate sets  $\mathcal{M}_t(\cdot)$  by (C1)–(C5), then compute  $\hat{\mu}$  by (7) and (8). Update the no-progress counter  $\nu_t$  by (20).
2. **Consider declaring.** For each live half-suit form  $\alpha^*$  by (18) and its probability  $\pi_{\text{alloc}}^*$  by (6). Declare if  $\Delta > \delta$  in (19), or if a pressure or forced-endgame override fires (§6.4).
3. **Otherwise ask.** Enumerate  $\mathcal{A}_t$  by (9) and score every candidate by (10), including the contestation term (17).
4. **Resolve ties.** Form  $\mathcal{E}_t$  by (15). If the endgame condition does not hold, return (16) and stop.
5. **Search.** Otherwise sample  $D_{\text{det}}$  worlds with weights (21), roll out each candidate in each world, and return the candidate selected by (23), which is the blueprint’s own choice unless a candidate clears it by  $\kappa$  standard errors.

Every step reads only the acting seat’s own hand, the public record and, in step 5, its own reset seed. That is what makes the coordination argument of §6.8 go through, and it is the property the side-channel audit of §7.10 tests mechanically rather than by inspection.

## 7 Evaluation framework

Roughly half of what this project has produced is its evaluation method, and it is presented here as method rather than as process, because each component was adopted in response to a specific measured failure and each is portable to another game. §4.3 describes the failures.

## 7.1 The outcome measure

The primary outcome is the team win rate, the proportion of games in which the evaluated configuration’s team takes at least five of the nine half-suits. Comparisons are reported as an **edge in percentage points**,

$$\text{edge} = 100 \times (\text{win rate} - 0.5),$$

so that zero denotes indistinguishable configurations and the sign gives the direction. The paper writes *pp* for percentage points of this quantity. Three other quantities appear and are never conflated with it: the mean number of half-suits a team takes, used for the search’s internal return; predictive log loss and argmax hit rate, used to score posteriors as predictors (§4.3); and the adversarial edge of §11.8, which is an edge in the same units but measured for a fitted opponent against the evaluated configuration.

## 7.2 Duplicate deal blocks

A *cell* compares two configurations *A* and *B* over a *bank* of deals identified by one integer seed. Every deal in the bank is played twice: once with *A* in the odd seats and *B* in the even seats, and once with the seats exchanged. Both configurations therefore play the identical deal from both sides.

Fish has large per-deal variance, since a deal can be won or lost on the cards, and an unpaired comparison spends most of its sample resolving that variance rather than the difference between policies. The rotation cancels the deal’s intrinsic advantage exactly, in the spirit of duplicate bridge and of the variance-reduction estimators developed for poker [64, 65]. The design requires reproducibility from the seed alone (§5.2); without it the two rotations are not the same deal and the design degrades silently into an unpaired one.

## 7.3 Deal-clustered bootstrap confidence intervals

The two rotations of one deal are not independent observations. They share a deal and their outcomes are negatively correlated by construction, so treating them as independent understates uncertainty.

Every interval in this paper is a two-sided 95% **deal-clustered bootstrap confidence interval**. The resampling unit is the *deal*, carrying both of its rotations together; 20,000 bootstrap resamples are drawn with replacement from the deals of the cell, the edge is recomputed on each resample, and the interval is the empirical 2.5th and 97.5th percentiles of that distribution. No Wilson or normal-approximation interval on the game count is quoted anywhere in this paper, and the substitution is not cosmetic: at these sample sizes it is the difference between a claim that replicates and one that does not.

Two derived quantities are used throughout and are defined once here.

**Pooling two banks.** Two banks of equal size pool as the mean of their two edges, with standard error  $\text{se}_{\text{pooled}} = \sqrt{\text{se}_1^2 + \text{se}_2^2}/2$ , where each *se* is recovered from that bank’s clustered interval as half-width divided by 1.96. The pooled interval is the estimate plus or minus  $1.96 \text{se}_{\text{pooled}}$ . Both per-bank values are printed beside every pooled figure in this paper.

**Differences between separately measured cells.** A difference between two pooled cells, such as an ablation’s leave-one-out drop, is the difference of the estimates with the two pooled half-widths combined in quadrature. This is conservative. The configurations are paired on deals, so a



genuine paired difference would be narrower, but the harness produces no paired difference *across* two separately executed cells, and constructing one would assert more than the data structure supports.

## 7.4 Power computed before a cell is run

At parity a cell’s 95% half-width in percentage points is approximately  $98/\sqrt{N}$  for  $N$  games. The harness computes this for every cell and emits it with the result. The arithmetic is applied in advance because it determines what may be attempted: resolving one point requires roughly 9,604 games, and a quarter of a point requires sixteen times that.

The failure this prevents is the one §4.3 describes, in which five mechanisms cleared a 95% paired interval at one budget and returned exactly zero at three times it. Sample sizes in this paper are chosen from the arithmetic before a cell is run, never after seeing its result.

## 7.5 Replication across disjoint banks as an advance-specified criterion

Every claim is measured on two disjoint deal banks, and the criterion was fixed before the data were seen:

A claim whose **sign** does not agree on both banks is reported as not replicated, whatever its pooled interval says.

This is stricter than a pooled significance test, deliberately. A pooled interval can exclude zero while the two banks disagree, and that configuration has occurred in this corpus and has previously been published as a positive result. The criterion binds against this paper’s own findings in §11.4 and again in §11.3, and both instances are reported.

## 7.6 Sealed material and advance registration

A programme that generates candidate configurations, scores them and selects the best is performing a maximisation, and the selected configuration’s score is upward biased by the amount a maximum over  $K$  draws exceeds the mean. No care taken in the final measurement removes that bias if the final measurement uses material the selection has seen.

Before any candidate existed, deal banks were reserved by seed and their digests committed, a digest being a rolling hash over the six dealt hands and the dealer of every deal in the bank, computed without constructing a policy or playing a game. An adversary population was split in half by a rule fixed in advance, and one half was encrypted with the plaintext’s SHA-256 committed, so that it could not be trained against even accidentally. The engine enforces the seal: a run naming a sealed seed fails unless an explicit environment variable is set, which was set once, at the start of the final evaluation, and recorded.

The entire final battery, comprising every cell, its sample size, its threshold, the arithmetic behind the sample size and the conditions that would count as non-confirmation, was written down and committed before any holdout bank had been played and before the sealed half had been decoded. The evaluation was then executed from a cleared context, reading that document and nothing else from the project’s records. Where it required a training figure for comparison it took it from the registered protocol, which states such expectations in advance for exactly this purpose, so that agreement constitutes a check and disagreement constitutes a finding.

Anything the evaluation did that the protocol does not specify is recorded as a deviation, next to the affected result. There are 27, listed in Appendix F.

## 7.7 Selection effects and multiple comparisons

This study evaluates many candidate mechanisms, opponents, ablations, adversarial searches and configurations, and the inferential status of that needs stating precisely.

**No formal family-wise or false-discovery correction was applied**, and no interval reported here is adjusted for multiplicity. What the design provides instead is structural. *Registration* fixes the primary comparison, its threshold and its interpretation before the data exist, so the primary claim is a single pre-specified test rather than the best of many. *Sealing* makes the holdout material unavailable to the selection process by construction rather than by discipline. *Disjoint banks with a sign criterion* require a claim to appear twice before it is reported. *Calibration* of the detection floor (§7.9) establishes what the instrument can resolve, so that a null is interpretable. And an explicit *selection check* (§11.2.3) computes how much of the measured gain a maximum over  $K$  candidates would produce under the null and compares advance-stated training figures against their holdout counterparts.

Exploratory comparisons in §9 are reported as exploratory: they were run on training material during the development phase, none is treated as confirmatory, and none is used to support the paper’s primary claim.

## 7.8 A soundness gate applied before any strength measurement

Win rate and soundness are dissociable in this policy family. The corpus contains a configuration that scores 56.60% against an earlier version while making 44.83% provably dead asks, asks whose target it can already prove does not hold the card (§6.1), and losing 14% of its games to the action cap. It is both stronger and broken.

A configuration is therefore gated *before* any strength number is computed for it, on eight rules covering dead asks, dead runs, action-limit games, the mirror game-length tail, late declarations, and the side-channel tests below. A configuration that fails is rejected regardless of its win rate. The gate runs on *training* material by design, being a soundness check on self-play behaviour rather than a strength claim. It is accompanied by a negative control, a configuration constructed to be rejected, because a gate that cannot reject it would certify nothing about the configuration it accepts.

## 7.9 Calibrated detection floors

An exploitability figure is produced by a search, and a search that finds nothing admits two explanations: there is nothing to find, or the search is too weak. A **detection floor** separates them. An artificial handicap of known size is planted in a target, a responder is fitted against the handicapped target, and the *excess* of that responder’s edge over its edge against the unhandicapped target is measured. The smallest planted edge the procedure recovers is the floor.

Two findings from that calibration shape every exploitability statement in this paper. First, the floor does not decrease with evaluation games. An early prediction that it would fall as (evaluation games)<sup>-1/2</sup> was refuted at its own predicted value: at four times the games the floor is 1.53 pp, and the sub-floor rung remains undetected not because the interval is wide but because the measured excess resolves to zero. Below roughly a point and a half a fitted responder is empty rather than merely unresolved, so the binding constraint is search power rather than evaluation power. Second, floors are family-specific: the declaration channel has its own and higher floor of 2.13 pp, and reading a declaration-objective result against the general floor would overstate it.

An adversary class contributes to an exploitability statement only after passing a planted-edge calibration at or below the effect size in question. That precondition is why the negative controls

of §11.9 are run: an instrument that cannot recover a planted edge of a given size cannot report the absence of a real one of that size.

## 7.10 Mechanical homogeneity and side-channel controls

The evaluated team is three identical agents with no shared secret. That their coordination is common knowledge is legitimate (§2.3), but it also means an implementation detail could become an unintended signalling channel. Before this cycle the constraint had only ever been checked by reading the source.

Four mechanical tests now run over all four engine decision types. **S3, listening substitution** asks whether a seat’s behaviour changes more when a teammate’s hidden action changes than when an opponent’s does, beyond what the rules explain. **S4, stream independence** asks whether the seat’s private random stream is drawn independently of the deal. **S5, posterior invariance** asks whether the seat’s action depends on true card locations beyond what its own posterior supports. **S6, seat isolation** replays every decision from own hand, public event stream, rules and reset seed alone, and counts the decisions the reconstruction cannot reproduce, at zero tolerance.

The tests are calibrated by three positive controls, each a policy carrying one planted violation, and each must fail exactly the tests the protocol names for it in advance and no others. §10.2 reports the calibration on holdout material. Without those controls a clean report from the gate would be uninformative.

One limitation is stated wherever the gate is used. The zero-tolerance form of S6 requires a single worker thread, because the audit depends on the deal-to-thread partition and work stealing does not fix that partition at higher counts, and it requires agents rebuilt for each deal. Rebuilding agents per deal *changes play*. The gate therefore certifies that with cross-deal residue removed, nothing else depends on anything it should not. The residue itself is not removed from the deployed mode, and every strength number in this paper is measured under it (§11.9).

## 7.11 Worst case and minimax regret over a shared panel

The project’s standing reporting rule is that the worst case leads and an aggregate never does, since a mean over an opponent panel rewards beating weak opponents by larger margins, which is not the quantity of interest. Two statistics are reported over a panel scored identically for every configuration: the **worst cell**, being the panel member on which a configuration does least well, and the **minimax regret**, being for each member the shortfall from the best configuration on that member, maximised over members. They can disagree, and in §11.3 they do. Reporting both, and reporting the panel mean last and labelled as a diagnostic, is the discipline.

Every configuration must carry a cell for every member or minimax regret is undefined, which is why the panel is fixed by name in advance and scored for all configurations on the same banks. One consequence of that design is recorded where it matters: a configuration that is itself a panel member meets itself and carries a free zero cell, which can only flatter its worst case, so §11.3 prints the worst case both with and without self-cells.

## 8 A threat model for a homogeneous deterministic team

Robustness claims are only meaningful against a stated adversary, and the adversary appropriate to this domain is unusual enough to define explicitly. The definition below was written before any v0.7 measurement was taken, and it constrains how §11.8 may be read.

**The unit is the team.** What is evaluated is the joint policy  $\pi = (\pi_1, \pi_2, \pi_3)$  of three identical agents. The principal adversary controls all three opposing seats. A one-seat deviation column is reported alongside it on the same deals, because a three-seat upgrade and a one-seat upgrade are different quantities that this corpus has previously conflated (§11.5).

**The adversary is white-box, with an explicit exclusion list.** It is given the target’s source, its parameter vector, and unlimited offline access to it. It is not given the deal, the deal seed, any seat’s hand, or the engine’s confidence field. The exclusions are the point of the construction. Because the target team is three identical published deterministic agents, its coordination is common knowledge, which inverts the assumption in the team-correlation literature that the correlation device is private [53, 54]. The adversary class built to exploit that property is the transcript inverter of §11.8.1.

**Homogeneity is exactly two prohibitions.** The team receives no shared secret pre-play seed, and no seat may use an illegal side channel, defined mechanically by the four tests of §7.10 over all four engine decision types. Everything else a seat-conditioned policy can do is permitted.

**Three correlation regimes, and which one is principal.** The three adversary seats may be fitted independently, may be three copies of one vector, or may share a pre-play signal secret from the target team. The first is the *independent* regime, the second the *synchronised* regime, and the third the *ex-ante correlated* regime; the last is the only one whose result would bear on the target rather than on the search alone, and even there it remains a lower bound. It is the registered principal case. The synchronised regime is retained as the continuity column, because it is what earlier cycles of this corpus measured. §12.7 records which of the three this study actually reached.

**Every exploitability figure is a lower bound.** Each is reported as at least  $x$ , with the adversary class, the number of controlled seats, the correlation regime and the search budget attached. Computing the true value is intractable even white-box. The consequence is stated once and observed throughout: **a search that fails to find an exploit is evidence about the search, not a property of the target.** The v0.4 record in this corpus contains exactly that reading error, having reported a null result as a demonstration of robustness, and §E carries the correction rather than repeating it.

**A class contributes only after calibration.** An adversary class is admitted to an exploitability statement only once it has passed a planted-edge calibration at or below the effect size being claimed (§7.9).

**What the definition does not license.** It does not license the claim that a policy passing every test is unexploitable, and it does not license reading the opponent panel’s worst cell as an exploitability figure. The panel measures robustness across playstyles; the adversary search measures what a fitted responder achieves. They are different objects and this paper keeps them in different subsections.

## 9 Developing SESTINA v1.0

This section reports the development phases: what the instrument was shown to be capable of detecting, what an open adversary search found, and which candidate mechanisms were tested and

rejected. All measurements here are on *training* material and are exploratory in the sense of §7.7. None supports the paper’s primary claim, and every pooled interval reported in this section has a lower bound below the detection floor of 1.53 pp, so each is a measured effect rather than a confirmed one.

### 9.1 The v0.6 frontier, and the choice of comparison target

v0.6 deployed its policy with test-time search disabled, while also having measured that search working. The configurations available at the end of that cycle therefore form a *frontier* rather than a point, running from the deployed blueprint to progressively more expensive searching configurations.

This distinction drove the most consequential methodological choice in the programme. Beating the deployed policy is the easy comparison, and several unfitted single-switch deviations of it already do. **The comparison target was therefore fixed as F-cheap**, the cheapest configuration that is genuinely on the frontier, and §10.4 records that the choice was registered in advance. Table 3 defines every configuration this paper compares.

### 9.2 What the instrument can detect

The detection-floor procedure of §7.9 was built and calibrated in this phase, which measured the floor at 1.68 pp and predicted that it would fall as (evaluation games)<sup>−1/2</sup>. The phase of §9.3 re-measured the same rung at four times the games and got 1.53 pp, refuting the prediction; that is the first of the two findings §7.9 reports, the second being that floors are family-specific, with the declaration channel’s at 2.13 pp. Two further results shape later sections.

**Transcript inversion contracts the posterior, and is not an attack.** Because the target team is deterministic and published, an adversary can invert its transcript, asking for each observed ask which deals would have produced it and contracting the posterior accordingly. Measured, this contracts the deal posterior by approximately 2.0 bits per observed ask beyond the certificates the rules already supply. Converting those bits into play is worth +1.52 pp unfitted against the deployed policy.

It is nonetheless not an exploit. The excess tracks planted-edge size to within  $\pm 0.4$  pp on all eight rungs of the calibration ladder, which is the signature of a general strength gain rather than an exploitation instrument. And no linear handicap adds more than 0.07 bits per ask, so within this policy family there is no readability handicap available for a stochastic policy to purchase. A brief that proposed a stochastic policy conditional on the inverter succeeding was therefore closed on its antecedent.

**A cost figure the record had wrong.** The v0.6 paper priced the searching configuration at “three orders of magnitude” the blueprint, a ratio that mixed an all-threads timing with a single-thread one. Re-timed here on one basis, it is 242 $\times$ . But that figure describes the *unrestricted* search, which is not the frontier point anything was measured at, and phase 3 re-measured the endgame-restricted operating point at approximately 3.2 $\times$ . §11.10 corrects that correction in turn.

### 9.3 Open adversary generation against the frontier

The constraint on this phase was that the searches must not share a bias, since fifteen runs of one search constitute one run. Every exploiter search in the corpus before it had been the same search: a cross-entropy fit of a linear vector maximising win rate from one seed against one target. This

phase constructed six adversary classes in the synchronised regime, plus one fitted search in the ex-ante-correlated regime, and measured channel ceilings before fitting anything against them.

**The principal finding is negative.** Nothing constructed exploited the v0.6 frontier beyond the detection floor. The best measured exploitability of the deployed policy remains an in-class figure of +0.79 pp [+0.48, +1.10] over 96,000 games, which is below the class floor of 1.53. Seven independently fitted searches confirmed it and none cleared its class floor.

Every configuration that beat the deployed policy by more than the floor proved on measurement to be a better policy rather than an exploit: the target’s own test-time search at +1.89 pp, and a single out-of-class coordinate at +2.71 pp whose gain survives against opponents sharing none of the target’s machinery. The second of those is the contestation weight of §6.3; §11.4 measures on holdout what it contributes and finds it the largest of the five components SESTINA v1.0 carries, rather than the whole of the gain.

Two further results from this phase bear on SESTINA v1.0’s design. The declaration-urgency channel that the phase expected to be its headline has a measured ceiling of +0.38 pp [−0.06, +0.81] and is dead as an attack, but the same channel is worth +1.23 to +1.62 pp to a configuration that simply disables it: the adversary owns the smaller half of a two-point defect in the target. And the event-count cliff of §6.5 is unreachable by any adversary, while a configuration that removes the counter inherits a self-play game-length tail of 405 events, which is the obligation §6.6 discharges.

## 9.4 Candidate mechanisms that were tested and rejected

Five candidate mechanisms were developed in isolated branches, gated before being scored, and checked against the side-channel definition by the harness of §7.10 rather than by inspection. **One survived**, and it is the configuration change described in §6 rather than an architecture. The four that were rejected are reported here, each with the measurement that rejected it, because each closes a standing hypothesis in the project’s design register and because a closed direction does not depreciate.

### 9.4.1 The declaration-allocation channel is worth zero

The design register’s second-ranked hypothesis, at priority 0.48, held that v0.6’s entire margin lives in the declaration channel and that roughly two points of it remained. Its proposed mechanism was that the deployed allocator, a product of per-card marginals, selects a different allocation from the joint posterior’s argmax often enough to matter.

It does not. Over approximately 210,000 recorded voluntary declarations, across two banks and both urgency settings, the joint argmax and the marginal-product argmax select the same allocation at every one, with the joint score holding a strict opinion on 95.6% to 100% of genuinely ambiguous cases. The reason is algebraic rather than statistical: the sequential joint’s first chain-rule factor *is* the marginal, and the renormalisation that follows reweights the survivors without reordering them.

Two further measurements close the direction rather than merely the mechanism. First, the exact joint maximiser is a worse decision rule than the deployed approximation on the declaration decision itself: paired, it corrects 218 declarations and breaks 454, a net −1.313 pp, which converts to approximately −1.58 pp of win rate. This is the second independent instance of the methodological rule from §4.3 paying off. Second, 72.4% to 74.4% of the error class occupies flat-posterior states in which the exact object is provably a coin flip, so most of the apparent headroom is an information limit rather than a policy defect.



The channel is also largely downstream of a defect that SESTINA v1.0 removes for other reasons. Disabling the urgency escalation eliminates 71% of the entire allocation-error class and drops the oracle ceiling for a perfect allocator from 2.45 to 0.84 pp, which is below the detection floor. The consequence is a composition constraint this paper observes: a declaration-channel gain and the urgency-off gain may never be added, because they are the same defect counted twice.

#### 9.4.2 A leaf evaluator fitted 74 times better buys nothing

An inherited hypothesis held that the test-time search cannot be judged until its leaf evaluator is rebuilt, the deployed one being algebraically close to a rescaling of the hit probability. A leaf was fitted for the purpose, and the statistic that matters, between-candidate  $R^2$ , which is the only component the paired deviation rule can consume, improved from 0.00108 to 0.07976 in sample, a 74-fold gain, and from 0.00664 to 0.08424 out of sample, so the direction replicates and the multiple does not. The fit was scored over approximately 1.3M leaves across two search regimes and three banks.

In play it is worth +1.84 pp against the material control’s +2.01 pp, losing on both banks individually. The diagnosis is conversion failure rather than fit failure: the rule’s deviation rate is 0.3205 for the deployed material leaf, 0.3217 for the fitted one and 0.3237 for a third arm that merely rescales the material leaf’s constant, so at the deployed shrinkage the deviation rule is nearly leaf-invariant. A better leaf is not the available lever. This reproduces, in a new domain, the abstraction pathology the poker literature documents [66]: a model improved on its own metric can leave play unchanged or worse.

The inherited hypothesis was discharged separately and by re-measurement rather than by the new evaluator. The single underpowered cell on which it rested, one 4,000-game cell at +0.08 pp, is the `depth=24` configuration; re-run on the training banks with v0.6’s own leaf it gives +1.02 [−0.55, +2.62], and full-game search at `depth=12` gives +1.78 and +1.25 pp, pooled +1.52. Phase 1’s +2.19 pp refuted the conditional in the endgame regime only, at `depth=12` with `maxq=26`; it was this full-game cell that left it binding for full-game search.

#### 9.4.3 Fitting against a per-decision objective produces a worse policy

A further hypothesis held that a per-decision objective is necessary because the scoreboard is too noisy to resolve small effects. Its precision claim is confirmed and its fitting form is rejected, and the two halves are independent.

The design effect of deal clustering was measured for the first time in this cycle, and a one-point-equivalent effect resolves in 284 to 560 games on the declaration channel and 244 to 299 on the allocation-error channel, against 9,604 on the scoreboard, confirming the precision claim at 17 to 39 times.

But all three self-oriented per-decision objectives, fitted at matched budget against a per-game control, produced worse policies: −2.26, −1.59 and −1.20 pp, each replicated in sign and two of them clearing the floor as confirmed negatives. Each fit moved its own proxy in the intended direction and lost anyway. The declaration-oriented fit purchased +0.39 pp of declaration accuracy by paying −2.33 pp of ask accuracy, and the ask-oriented fit did the reverse. The two per-decision channels trade against each other, so a fitter holding 55 coordinates buys one out of the other. The estimator is worth keeping and the objective is not.

#### 9.4.4 Nothing learned survives a free random tie-break

In the v0.6 mirror 53.80% of contested ask decisions tie bit-for-bit, and the open question was whether anything can be *selected* within that set rather than merely randomised. The learned candidate this cycle built was not built for that question. It was the brief’s wildcard, aimed at amortising test-time search, and it was fitted unrestricted over 88,502 searched decisions. Deployed unrestricted it scores  $-1.83$  pp pooled over 48,000 games, negative on both banks  $[-2.22, -1.45]$ . Restricting it to the tied set is one added switch on the same fitted weight vector, and it is there that the tie-question gets answered.

Restricted, against the deployed policy it scores  $+1.19$  pp, indistinguishable from the free public-history tie-break’s  $+1.14$  pp; head to head against that free tie-break at 48,000 games it is  $-0.01$  pp  $\pm 0.45$ , replicated. Nothing learned survives the free tie-break, and the correct control for any mechanism operating inside the tied set is the decorrelated policy rather than the deployed one.

The same instrumentation explains why. Setting every candidate’s true advantage to exactly zero and feeding the capture’s real standard errors through the engine’s real deviation rule reproduces the search’s observed deviation rate to a 0.4% relative error, on a quantity the null was never fitted to. The search’s per-decision selection signal is approximately 84% winner’s curse on its own Monte Carlo draw, and approximately 95.5% within the tied set, where the rule applies no shrinkage at all.

#### 9.4.5 Two directions closed on arithmetic

Closing the entire forced-endgame accuracy gap, re-derived from this cycle’s own capture, is worth 0.030 pp, which is the register’s own figure corrected upward by a factor of two and still a fiftieth of the detection floor. The single route by which it could have returned is an adversary driving the target’s incidence upward, and the one construction measured does not do that: a team that never declares voluntarily raises its *own* forced-endgame rate, not its opponent’s, and pays 6 to 8 pp for it. No construction that raises the target’s incidence was measured, so the route is closed by the absence of a mechanism rather than by a price.

### 9.5 Selection and freezing

15 configurations were scored against the deployed policy during development. One was selected, its specification written to an artifact whose reconstruction is executed rather than asserted, and the configuration frozen. Nothing about it was changed afterwards, and no gradient or population fit was applied to it. §11.2.3 quantifies what that selection could be worth under the null.

## 10 Experimental design

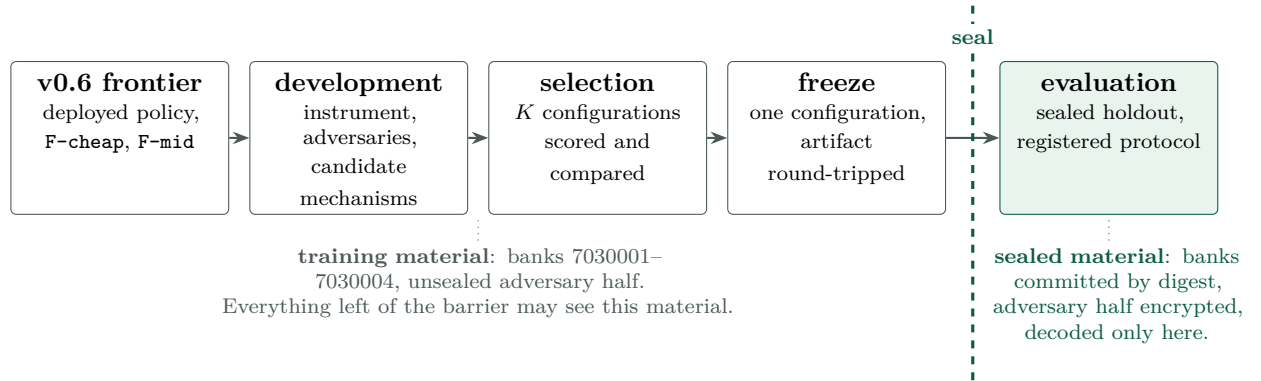
Figure 2 is the shape of what follows: what was developed on training material, what was frozen, and what was measured against sealed material.

### 10.1 Configurations compared

Table 3 is worth reading before any result, because the paper’s central nuance depends on the distinction between two of its rows. SESTINA v1.0 is compared against **F-cheap**, a configuration that existed before this cycle, and improves on it. It is also compared against the phase-2 composite, which was assembled *during* this cycle by the adversary-generation phase, and does not measurably improve on it. Those are different claims about different objects and §11.2.1 keeps them apart.

**Table 3:** Every configuration this paper compares. Configurations marked pre-existing were available before the v0.7 programme began; the phase-2 composite was assembled during it. Only SESTINA v1.0 was frozen for holdout evaluation.

| Name                  | What it is                                                          | Origin                           | Role in this paper                                                                            |
|-----------------------|---------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------|
| SESTINA v1.0          | the evaluated configuration of §6                                   | frozen in this cycle             | the subject; evaluated on sealed holdout material                                             |
| v0.6 deployed         | the previous release, search disabled                               | pre-existing                     | lineage baseline and reference opponent for ablations                                         |
| F-cheap               | cheapest configuration genuinely on the v0.6 frontier               | pre-existing                     | <b>the registered primary comparison target</b>                                               |
| F-mid                 | a more expensive frontier configuration                             | pre-existing                     | secondary frontier comparison                                                                 |
| the phase-2 composite | v0.6 frontier search, plus the contestation weight, plus one switch | assembled in development phase 2 | the strongest configuration available before the architecture work; the comparison of §11.2.1 |
| v0.5, v0.4, v0.3      | earlier lineage releases                                            | pre-existing                     | lineage baselines and panel members                                                           |
| scripted archetypes   | thirteen hand-built playstyles, including deception archetypes      | pre-existing                     | opponent panel members                                                                        |
| sealed adversaries    | fourteen configurations sealed before this cycle                    | sealed in phase 2                | opponent panel members, decoded only at evaluation                                            |



**Figure 2:** The development and evaluation protocol. Configurations were generated, scored and selected using training material only. Exactly one configuration was then frozen, and evaluated against material whose digests had been committed before any candidate existed, under a protocol registered in advance (§7.6). The seal is enforced by the engine rather than by convention: a run naming a sealed seed fails unless an explicit variable is set, which was set once and recorded.

## 10.2 Material, and the checks performed before any measurement

All 7 bank digests reproduce, at 24,000 deals each (Table 4). The sealed adversary half decodes to 14 configurations whose plaintext SHA-256 matches the commitment recorded at sealing. The frozen configuration’s artifact verifies, its specification rebuilding character for character from its own ordered option map, and its round trip with search enabled is identical over 800 games. The engine’s soundness audit reports 0 violations in 6,737,436 checks and its self-test passes on 424,986 checks.

**Table 4:** The deal banks. Each digest was recomputed before any cell was played, by folding the six dealt hands and the dealer of every deal into a rolling hash without constructing a policy or playing a game. The committed digests had been computed by a different build of the engine at sealing time, so agreement is between two binaries as well as two dates.

| Bank    | Deals  | Digest reproduced | Matches commitment |
|---------|--------|-------------------|--------------------|
| 7090001 | 24,000 | 896dbc89be124d85  | ✓                  |
| 7090002 | 24,000 | 0b6e40d834ac0ca1  | ✓                  |
| 7090003 | 24,000 | 863bea69baf6e73c  | ✓                  |
| 7090004 | 24,000 | 54f257c3f8ae9fab  | ✓                  |
| 7090005 | 24,000 | 268a1dae71a31713  | ✓                  |
| 7091001 | 24,000 | 958ada042cc26900  | ✓                  |
| 7091002 | 24,000 | 5c39af3b5e0bd9a0  | ✓                  |

**Table 5:** Calibration of the side-channel controls on holdout material. *Must fail* is the set the protocol names for each planted violation in advance. It is not one test each: the seed cheat is visible to two of them. A control failing a test outside its set, or missing one inside it, would mean the gate does not discriminate.

| Planted channel   | S3   | S4   | S5   | S6   | Must fail | As required |
|-------------------|------|------|------|------|-----------|-------------|
| v07x:cheat=seed   | PASS | FAIL | FAIL | PASS | S4, S5    | ✓           |
| v07x:cheat=shared | PASS | PASS | PASS | FAIL | S6        | ✓           |
| v07x:cheat=conv   | FAIL | PASS | PASS | PASS | S3        | ✓           |

The homogeneity controls of §7.10 calibrate as specified: each planted violation fails exactly the tests named for it and no others, identically on both banks, over 12 cells (Table 5).

## 10.3 The soundness gate

SESTINA v1.0 passes all eight rules of Table 6: 0.05820% provably dead asks against a 0.10% bar, longest dead run 1, mirror tail maximum 134 with 99th percentile 130, and 0 seat-isolation mismatches over 542,483 audited decisions. Reported but not gated: 99.805 events per game, 50.716% ask hit rate and 2.556% misdeclaration. The negative control fails 5 rules, namely G1, G2, G3, G4, G5, which are exactly the rules the protocol names for it in advance.

## 10.4 The registered protocol

The protocol fixed, before any holdout deal was played: the primary comparison target (**F-cheap**, for the reason in §9.1); the primary decision rule; seven secondary claims each with its own threshold; and seven conditions that would constitute non-confirmation. Appendix C reproduces them.

**Table 6:** The soundness gate, eight rules, run on training material by design. The negative control is a configuration constructed to be rejected.

| Rule                                             | SESTINA v1.0 | v0.6        | F-cheap     | Negative control |
|--------------------------------------------------|--------------|-------------|-------------|------------------|
| provably-dead asks $\leq 0.10\%$                 | ok           | ok          | ok          | <b>fail</b>      |
| longest dead run $\leq 5$                        | ok           | ok          | ok          | <b>fail</b>      |
| games with a run $\geq 6 = 0$                    | ok           | ok          | ok          | <b>fail</b>      |
| action-limit games = 0                           | ok           | ok          | ok          | <b>fail</b>      |
| mirror tail max < 220, p99 $\leq 150$            | ok           | ok          | ok          | <b>fail</b>      |
| declarations at/after event 220 = 0              | ok           | ok          | ok          | ok               |
| S3/S4/S5 certified, zero tolerance               | ok           | ok          | ok          | ok               |
| S6 = 0 at <code>--threads=1 --freshagents</code> | ok           | ok          | ok          | ok               |
| <b>verdict</b>                                   | <b>PASS</b>  | <b>PASS</b> | <b>PASS</b> | <b>FAIL</b>      |

The primary decision rule is worth stating in the body, because the paper avoids the word *certified* in favour of naming what was actually satisfied. The registered rule was that the comparison against **F-cheap** would be regarded as a confirmed advance if the pooled edge had a lower bound above 1.53 pp, *and* the sign replicated on both banks, *and* the configuration passed the soundness gate. A pooled edge positive with a lower bound below 1.53 but above zero, with the sign replicating, would be recorded as a measured but unconfirmed advance; an interval containing zero, a sign that did not replicate, or a gate failure would be recorded as no advance.

The protocol also fixed in advance the sentence to be used if the partner-regime comparison retained a particular ordering, so that its direction could not be chosen after the fact, and recorded that on training material that comparison ran *against* SESTINA v1.0. §11.5 reports the outcome and uses the registered wording.

## 10.5 The battery

428 scored cells over 4,322,400 games (Table 7). No registered cell was dropped; 18 were added, and Appendix F records why.

**Table 7:** Every registered battery against what was measured. *Registered* is the count the protocol names, not the count that ran: two deviations add a reference cell and sixteen control cells, because without them three of the protocol’s own questions are not computable from the cells it lists.

| Battery | What it measures                       | Registered | Added | Measured | Status        |
|---------|----------------------------------------|------------|-------|----------|---------------|
| B2      | headline strength against the frontier | 10         | 0     | 10       | complete      |
| B3      | the shared 31-member panel             | 248        | 0     | 248      | complete      |
| B4      | a fresh adversary search               | 16         | 0     | 16       | complete      |
| B5      | the attribution lattice                | 22         | 2     | 24       | complete (D1) |
| B6      | the partner-regime table               | 64         | 0     | 64       | complete      |
| B7      | cross-play between independent runs    | 24         | 0     | 24       | complete      |
| B8      | the rule-dialect table                 | 16         | 0     | 16       | complete      |
| B9      | the negative controls                  | 10         | 16    | 26       | complete (D2) |
| B10     | the S6 residual                        | 16         | 0     | 16       | complete      |

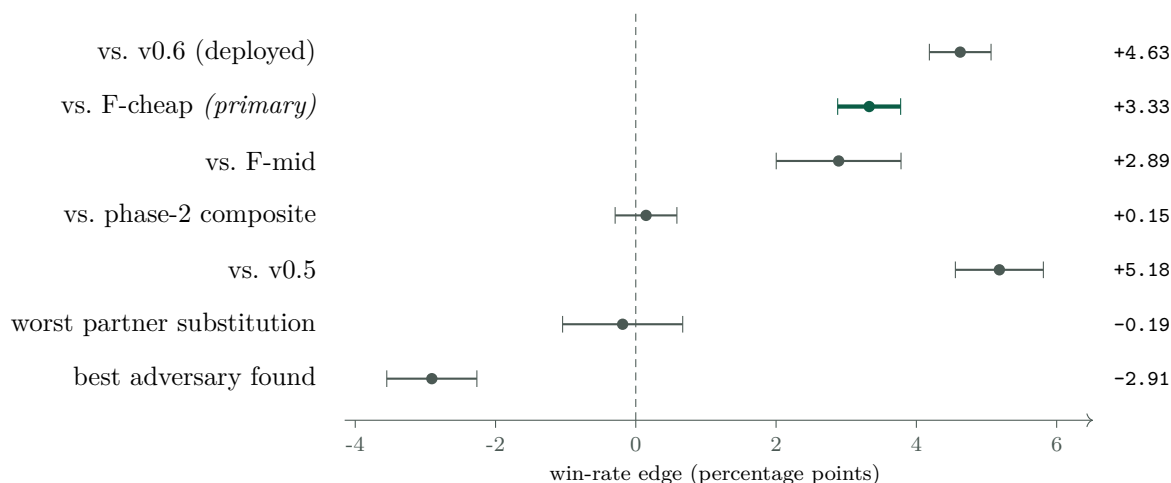
Scored cells carry no soundness-audit evidence, and this paper does not imply otherwise. The match harness does not run the audit, so the audit check count is 0 in all 428 of them, and a report of zero violations over a scored cell would be zero out of zero. The engine-wide audit figure quoted in §5.6 and §10.2 comes from the dedicated verification run and is not restated as a property of the battery.

## 11 Results

Every holdout measurement in this section is produced from the artifacts by the generator described in Appendix G and emitted into this document mechanically. The figures those measurements are read against are not measurements of this holdout: registered expectations, earlier cycles’ results, and structural counts. Appendix G records where each comes from. The section answers RQ1 in §11.1, RQ3 in §11.4, and RQ2 in §11.5 through §11.8, before reporting the validity controls and the computational cost.

### 11.1 RQ1: strength against the v0.6 frontier

Against **F-cheap**, the registered primary target, SESTINA v1.0 gains **3.33 pp** of win-rate edge  $[+2.88, +3.78]$  over 48,000 games, with the sign replicating on both banks ( $+3.67$  and  $+2.99$ ) and the lower bound above the registered 1.53 pp threshold. The configuration passed the soundness gate (§10.3). All three conditions of the registered primary decision rule are therefore satisfied, and the comparison satisfies the registered primary criterion. Figure 3 places that comparison and the others on one axis, and Table 8 gives the cells behind it.



**Figure 3:** Principal comparisons for SESTINA v1.0, pooled over both holdout banks with 95% deal-clustered bootstrap confidence intervals. Positive values favour SESTINA v1.0. The registered primary comparison is highlighted. The bottom two rows place two robustness results on the same axis: the worst partner substitution of §11.5, and the most successful of the 8 adversarial searches of §11.8.

Against the lineage baselines, SESTINA v1.0 gains  $+4.63$  pp over the deployed v0.6 policy  $[+4.19, +5.06]$ ,  $+2.89$  pp over **F-mid**  $[+2.00, +3.78]$ , and  $+5.18$  pp over v0.5.

**Against the phase-2 composite it gains  $+0.15$  pp  $[-0.29, +0.59]$ , an interval containing zero.** §11.2.1 treats that result, which the protocol had named in advance as a condition of non-confirmation.



**Table 8:** Strength against the v0.6 frontier and lineage baselines, paired duplicate blocks, both holdout banks.

| Cell | Opponent              | Edge (pp) | 95% CI         | Games  | Sign replicates |
|------|-----------------------|-----------|----------------|--------|-----------------|
| B2.1 | v0.6                  | +4.63     | [+4.19, +5.06] | 48,000 | yes             |
| B2.2 | F-cheap               | +3.33     | [+2.88, +3.78] | 48,000 | yes             |
| B2.3 | F-mid                 | +2.89     | [+2.00, +3.78] | 12,000 | yes             |
| B2.4 | the phase-2 composite | +0.15     | [−0.29, +0.59] | 48,000 | yes             |
| B2.5 | v0.5                  | +5.18     | [+4.56, +5.81] | 24,000 | yes             |

**Declaration accuracy is reported without a claim.** The declaration-family floor applies to the declaration-accuracy column. The harness emits declaration accuracy as a bare rate with no interval of any kind, so no interval can be quoted and no claim about the declaration channel is made. SESTINA v1.0 runs 0.970 to 0.982 across the five cells against opponents’ 0.976 to 0.986. That is a diagnostic.

## 11.2 Non-confirmatory results

The protocol named seven conditions in advance that would constitute non-confirmation, because after the data are in every one of them admits a reading that makes it sound like something else. Two arose, and they are reported here rather than in a discussion of caveats.

### 11.2.1 SESTINA v1.0 does not measurably outperform the phase-2 composite

SESTINA v1.0 against the phase-2 composite is +0.15 pp [−0.29, +0.59] over 48,000 games. Both banks are positive, so the sign replicates, but the interval contains zero.

The registered interpretation is explicit, and this paper adopts it. The increment that an earlier phase had reported over the composite, +1.42 [+0.18, +2.68] on a single bank, did not replicate; on this comparison SESTINA v1.0 reduces to a configuration that development phase 2 had already assembled.

The arithmetic makes the cell interpretable rather than merely disappointing. SESTINA v1.0 minus the phase-2 composite is exactly the tie-break repair, the urgency removal and the stall detector added, with one further switch removed. §11.4 measures the stall detector’s contribution as identically zero, and the removed switch’s leave-one-out drop was zero on training material. The protocol states that group’s training value in advance as +0.78 [+0.33, +1.22]; on holdout it is +0.15 [−0.29, +0.59].

What this does and does not license is worth stating precisely, because F-cheap and the phase-2 composite are different objects (Table 3). It does not retract §11.1: F-cheap is a configuration of the v0.6 frontier that existed before this cycle, and SESTINA v1.0 improves on it by a replicated margin. What it licenses is the narrower statement that **the strength gain in this cycle is real and was produced in development phase 2, and the architecture work that followed added no measurable strength on top of it.**

### 11.2.2 The attribution location test does not replicate

The protocol’s third non-confirmation condition is that the lattice cannot locate the gain: if removing every single component in turn costs less than a third of the whole, the configuration has a gain it cannot explain. Read per bank, as §7.5 requires, the two banks disagree (Table 9).

**Table 9:** The attribution location test, read per bank. *Whole* is the measured whole, *one third* the protocol’s threshold, and *largest single drop* the biggest leave-one-out drop the lattice contains.

| Bank    | Whole | One third | Largest single drop | Verdict     |
|---------|-------|-----------|---------------------|-------------|
| 7090001 | 4.883 | 1.628     | L-r12 +2.350        | LOCATED     |
| 7090003 | 4.575 | 1.525     | L-r12 +1.025        | NOT LOCATED |

The two banks’ largest drops differ by 1.325 pp against a half-width on that difference of comparable size, so the split is real on the point estimates and is not itself statistically resolved. Under the replication criterion the pooled verdict is not replicated (no), and the honest reading is that this battery does not settle whether the gain is located by the protocol’s one-third rule. Note that the disagreement is between verdict labels rather than between signs, which is itself a recorded extension of the criterion. The sub-additivity claim of §11.4 passes on both banks; it is the location test specifically that splits.

### 11.2.3 The selection check

15 configurations were scored against the deployed policy during development. At the lattice cell size the per-cell standard deviation is 0.3163 pp, so the quantity the protocol registers as the expected maximum of 15 draws under the null that none differs from the reference is  $\sigma\sqrt{2\ln K} = 0.74$  pp, which is an upper bound on that expectation rather than the expectation itself.

The selection term on holdout material is 0.00 pp *by construction*: the holdout banks were never available for selection, so not one of the 15 choices could have used any deal of them. That is a structural guarantee rather than a measurement. Table 10 sets the four registered quantities against their holdout measurements.

**Table 10:** The four quantities the registered selection check tabulates, against their holdout measurements. Three reproduce or exceed; one falls short. These are not every figure the protocol states in advance, and two of the partner-regime predictions in §11.5 also come in low.

| Quantity                                                     | Training | Holdout | Shortfall |
|--------------------------------------------------------------|----------|---------|-----------|
| B2.4 — <b>rtie</b> + urgency-off + <b>stall</b> , as a group | +0.78    | +0.15   | −0.63     |
| B6 self-play row, opponent v05                               | +2.94    | +4.49   | +1.55     |
| B7 diagonal (self-play)                                      | +4.51    | +4.55   | +0.04     |
| B7 off-diagonal (cross-play)                                 | +4.48    | +4.55   | +0.07     |

3 of the four reproduce or exceed their training values. The largest shortfall, 0.63 pp against the registered 0.74, falls on the composite cell, which is precisely the quantity measuring the group of components the configuration was selected over. That is the pattern the check was written to look for and it appears where selection occurred. One quantity cannot establish that the group’s training value was entirely selection effect, and the check is not a formal correction (§7.7); what it provides is a consistency test that the data pass in the direction §11.2.1 indicates.

### 11.2.4 The seven conditions, answered

Table 11 answers each of the seven in the protocol’s own order.

Two of the seven are met or unresolved, being condition 2 outright and condition 3 by non-replication. The other five are not met.

**Table 11:** The seven registered non-confirmation conditions against what the holdout returned. Appendix C reproduces the conditions verbatim.

| # | Condition                                      | Result                                                                   |
|---|------------------------------------------------|--------------------------------------------------------------------------|
| 1 | the frontier comparison                        | not met. +3.33 [+2.88, +3.78], sign replicates                           |
| 2 | the composite comparison                       | <b>met.</b> +0.15 [−0.29, +0.59]                                         |
| 3 | attribution: the location test                 | <b>does not replicate.</b> LOCATED on one bank, NOT LOCATED on the other |
| 4 | brittleness under partner change or cross-play | not met. Minimum −0.19; cross-play off-diagonal minus diagonal +0.01     |
| 5 | a successful adversary                         | not met. No search found a positive edge; largest upper bound −2.27      |
| 6 | the soundness gate                             | not met. All 8 rules pass                                                |
| 7 | the validity controls                          | not met. Every control behaves as specified                              |

### 11.3 Worst case and minimax regret over a shared panel

248 cells. Four configurations were scored against the same 31-member panel on the same two banks: 19 near-class members at 6,000 deals per bank, 10 far-class archetypes at 1,500, and 2 expensive members at 1,200. All four carry a cell for every member, which is what makes minimax regret defined. Table 12 gives each configuration’s worst cell.

**Table 12:** Worst panel cell per configuration. Per-bank values are given in bank order, 7090001 then 7090002.

| Configuration     | Worst cell            | Edge (pp) | 95% CI         | Per bank      |
|-------------------|-----------------------|-----------|----------------|---------------|
| SESTINA v1.0      | phase-2 composite     | −0.04     | [−1.41, +1.33] | +1.67 / −1.75 |
| v0.6 deployed     | <b>SEALED:X01xC3f</b> | −4.24     | [−4.86, −3.62] | −3.86 / −4.63 |
| <b>F-cheap</b>    | phase-2 composite     | −2.08     | [−3.48, −0.69] | −2.21 / −1.96 |
| phase-2 composite | phase-2 composite     | +0.00     | [+0.00, +0.00] | +0.00 / +0.00 |

SESTINA v1.0’s worst cell is −0.04 pp [−1.41, +1.33] against the deployed policy’s −4.24 [−4.86, −3.62], satisfying the registered secondary claim that the worst case not be catastrophic.

Two qualifications belong with that number, and both go against SESTINA v1.0.

First, **it does not replicate in sign.** SESTINA v1.0’s worst cell is +1.62 on the first bank and −1.75 on the second, attained at a different panel member on each (**SEALED:X01xC3f** and **composite**); on one bank SESTINA v1.0 has no negative cell anywhere on the panel. Under the criterion of §7.5 the pooled value is not replicated. What survives replication is the comparison the protocol actually gated on, since SESTINA v1.0’s worst cell is far above the deployed policy’s on both banks, rather than the pooled value itself.

Second, **it is not the best worst case of the four.** Configurations that are themselves panel members carry a free zero cell against themselves, so the worst case is printed both ways; the phase-2 composite is ahead of SESTINA v1.0 on both readings, at +0.00 with its self-cell and +1.23 without.

SESTINA v1.0 is **3rd of four** on minimax regret (Table 13), at 4.53 against F-cheap’s 4.00 and the phase-2 composite’s 4.08. The configuration with the best worst case is not the configuration with the best regret, and this paper does not resolve the tension by selecting whichever statistic favours the subject. The regret difference is concentrated in the far archetypes (Table 14), opponents

**Table 13:** Minimax regret over the shared panel, best first. The protocol sets no threshold on this quantity and requires only that it be reported.

| Configuration     | Minimax regret (pp) | Attained against      |
|-------------------|---------------------|-----------------------|
| <b>F-cheap</b>    | 4.00                | v0.3                  |
| phase-2 composite | 4.08                | <b>withholder</b>     |
| SESTINA v1.0      | 4.53                | <b>withholder</b>     |
| v0.6 deployed     | 5.67                | <b>SEALED:X01xC3f</b> |

every configuration already beats by between 24.72 and 50.00 pp, which is a reason to weight it less but not a reason to discount it.

**Table 14:** The far archetypes, where the minimax-regret differences are concentrated.

| Far archetype      | SESTINA v1.0 | v0.6   | <b>F-cheap</b> | Composite |
|--------------------|--------------|--------|----------------|-----------|
| v0.3               | +28.72       | +25.47 | +24.72         | +28.40    |
| v0.2               | +34.32       | +31.68 | +32.67         | +33.17    |
| <b>lockout</b>     | +30.78       | +28.65 | +28.97         | +29.07    |
| <b>detective</b>   | +29.65       | +28.08 | +27.98         | +28.32    |
| <b>hunter</b>      | +46.93       | +48.58 | +48.65         | +47.10    |
| <b>diversifier</b> | +43.40       | +44.90 | +45.18         | +43.32    |
| <b>bluffer</b>     | +49.87       | +49.98 | +49.95         | +49.88    |
| <b>silent</b>      | +32.75       | +34.23 | +34.05         | +30.92    |
| <b>withholder</b>  | +25.35       | +29.88 | +29.55         | +25.80    |
| <b>random</b>      | +50.00       | +50.00 | +50.00         | +50.00    |

Against the hardest sealed adversary the ordering reverses: on **SEALED:X01xC3f**, SESTINA v1.0 scores +1.43 [+0.80, +2.06] where the deployed policy scores −4.24 and **F-cheap** scores −1.25. 2 of the four configurations beat that member with a lower bound above zero, namely **FROZEN**, composite.

**The panel mean, reported last and as a diagnostic.** Over the 31 shared cells, which comprise 28 distinct policies since three are named twice and are therefore double-counted by any mean, the figures are SESTINA v1.0 +14.94, the phase-2 composite +14.35, **F-cheap** +13.28 and the deployed policy +11.99. The reporting rule of §7.11 forbids leading with this, and it points the opposite way from minimax regret. Appendix D gives every cell.

## 11.4 RQ3: attribution of the gain

The registered prediction was that the contestation weight of §6.3 carries the gain, checkable as a leave-one-out drop of roughly two percentage points. On that check it holds: the holdout drop is +1.69 pp [+0.80, +2.57], with the training figure +2.10 inside the interval, and an add-one-in of +2.18. The stronger statement, that this coordinate is where the gain lives, is not supported at this power: the drop’s lower bound is +0.80, and §11.2.2 reports the protocol’s own location test failing to replicate. Table 15 is the lattice.

**Table 15:** The attribution lattice. Add-one-in is measured from the deployed v0.6 policy; leave-one-out is the drop from SESTINA v1.0. Reference cell SESTINA v1.0 against the deployed policy on the lattice banks: +4.73 [+4.10, +5.36].

| Component                                | Add-one-in | 95% CI         | Leave-one-out | 95% CI         |
|------------------------------------------|------------|----------------|---------------|----------------|
| <b>r12=25</b>                            | +2.18      | [+1.55, +2.81] | +1.69         | [+0.80, +2.57] |
| <b>the search</b>                        | +1.91      | [+1.38, +2.44] | +0.57         | [−0.33, +1.46] |
| <b>rtie=1</b>                            | +0.71      | [+0.08, +1.34] | +1.08         | [+0.20, +1.97] |
| <b>urgency-off</b>                       | +1.37      | [+1.12, +1.63] | +0.64         | [−0.25, +1.53] |
| <b>stall=12</b>                          | +0.00      | [+0.00, +0.00] | +0.00         | [−0.89, +0.89] |
| <b>m2=0</b> ( <i>not in the freeze</i> ) | +0.77      | [+0.66, +0.88] | —             | —              |

The stall detector behaves exactly as predicted. It is worth +0.00 [+0.00, +0.00] added and +0.00 removed, bit-identical on holdout in win rate and in events per game alike: the rule does not fire.

The naive sum of the five components SESTINA v1.0 carries is +6.18 pp against a measured whole of +4.73 pp, a sub-additivity ratio of 0.766 [0.599, 0.933], which contains the phase-2 composition figure of 0.83. Quoting the sum would overstate the configuration by 1.45 pp. The registered attribution claim, that components are sub-additive and no single leave-one-out drop exceeds the whole, is satisfied.

One component was measured but is not carried by SESTINA v1.0. The **m2** switch is worth +0.77 [+0.66, +0.88] added to the blueprint, an interval excluding zero and larger than the tie-break’s add-one-in. Its zero leave-one-out drop was measured on training material and is not re-measured here, because SESTINA v1.0 does not carry the switch and the protocol directs that it not be added back; there is no leave-one-out cell for it and there could not be.

## 11.5 RQ2: partner substitution and one-seat upgrades

6 of 7 changed-partner rows against a v0.5 opponent are positive (Table 16). The minimum is −0.19 [−1.04, +0.67], which is above the registered −1.00 collapse threshold on the point estimate but *not resolved against it*, since the interval’s lower bound falls below the threshold. This battery does not separate a pass from a failure on the row that decides the claim, and reading a threshold from a point estimate is a move this paper declines elsewhere (§11.9) and declines here.

**Table 16:** Partner-regime table against a v0.5 opponent. Team A is the evaluated configuration plus two copies of the named partner; team B is three copies of the opponent. Per-bank values are given in bank order, 7090001 then 7090002.

| Partners          | SESTINA − v0.6 | 95% CI         | Per bank      | Repl.     |
|-------------------|----------------|----------------|---------------|-----------|
| itself            | +4.49          | [+3.61, +5.38] | +4.18 / +4.81 | yes       |
| v0.6              | +1.93          | [+1.04, +2.82] | +2.07 / +1.79 | yes       |
| v0.5              | +0.78          | [−0.09, +1.66] | +0.95 / +0.62 | yes       |
| v0.4              | +1.34          | [+0.46, +2.22] | +1.30 / +1.38 | yes       |
| v0.3              | +1.40          | [+0.56, +2.25] | +1.08 / +1.72 | yes       |
| <b>detective</b>  | +1.83          | [+0.97, +2.68] | +1.35 / +2.30 | yes       |
| <b>withholder</b> | −0.19          | [−1.04, +0.67] | +0.00 / −0.38 | <b>no</b> |
| <b>lockout</b>    | +1.89          | [+1.04, +2.74] | +1.39 / +2.39 | yes       |

Against a v0.6 opponent all four rows are positive and clear zero (Table 17), and this is where the measurement disagrees with the registered expectation. The protocol states in advance +4.82 in self-play and +2.50 to +2.88 under partner change. Measured: +4.55 in self-play, and of the three changed rows two fall below the stated range, namely +1.46 against v03, more than a point below its floor, and +2.34 against **detective**.

**Table 17:** Partner-regime table against a v0.6 opponent. Per-bank values are given in bank order, 7090001 then 7090002.

| Partners         | SESTINA – v0.6 | 95% CI         | Per bank      | Repl. |
|------------------|----------------|----------------|---------------|-------|
| itself           | +4.55          | [+3.92, +5.18] | +4.88 / +4.21 | yes   |
| v0.6             | +2.75          | [+2.14, +3.37] | +2.97 / +2.54 | yes   |
| v0.3             | +1.46          | [+0.63, +2.29] | +1.25 / +1.67 | yes   |
| <b>detective</b> | +2.34          | [+1.49, +3.19] | +2.42 / +2.26 | yes   |

**Table 18:** Transfer across partners: SESTINA v1.0 over v0.6 against v0.6 over v0.5, on the same eight settings and the same banks. The two training rows are quoted from the registered protocol; the protocol records that its v0.6-over-v0.5 training row is computed on three of seven changed settings, so that row is not directly comparable with the holdout row above it. Reported and not gated.

| Comparison                                      | Min   | Median | Self  | Median/self |
|-------------------------------------------------|-------|--------|-------|-------------|
| SESTINA over v0.6 — holdout                     | −0.19 | +1.40  | +4.49 | 0.313       |
| v0.6 over v0.5 — holdout                        | −0.40 | +0.26  | +0.69 | 0.380       |
| SESTINA over v0.6 — training, stated in advance | −0.15 | +1.26  | +2.94 | 0.428       |
| v0.6 over v0.5 — training, stated in advance    | −0.61 | +0.64  | +1.35 | 0.472       |

The protocol stated in advance that if the ordering in Table 18 survived, the appropriate summary is that SESTINA v1.0’s advantage transfers across partners about as well as v0.6’s did and, on the median-to-self-play ratio, slightly less well. The ordering survived (0.313 against 0.380), so that is the paper’s summary. The fact it omits is that SESTINA v1.0’s absolute changed-partner median is +1.40 against v0.6’s +0.26, and the ratio is smaller mainly because the self-play row it divides by is more than six times larger.

A one-seat upgrade is worth +1.93 pp against a v0.5 opponent (registered expectation +1.26) and +2.75 pp against a v0.6 opponent (registered +2.88). That a single seat purchases a large fraction of the three-seat gain is the strongest evidence in this paper that the advantage is individual rather than conventional.

## 11.6 RQ2: cross-play between independently trained runs

Diagonal mean +4.55, off-diagonal mean +4.55, off-diagonal minus diagonal +0.01 pp against a per-cell half-width of 0.63. The worst off-diagonal cell (+3.80) is 0.40 pp below the weakest diagonal cell (+4.20), and the best off-diagonal (+5.15) exceeds every diagonal cell. The registered claim that the advantage is not a private convention between identical fits is satisfied against a threshold of 1.50 pp. For scale, the coordination literature’s documented convention collapses run from self-play scores in the twenties to low single digits under cross-play [39, 42]; nothing of that kind is present. Table 19 is the cross-play matrix and Table 20 the three pairs head to head.



**Table 19:** Cross-play. Three runs of the same architecture, trained independently on disjoint fitting banks with different search trajectories and, for one, a different starting basin; pairwise parameter distance 6.3 to 9.6 over 55 coordinates.

| Run \ partners | xp1                  | xp2                  | xp3                  |
|----------------|----------------------|----------------------|----------------------|
| xp1            | +4.54 [+3.91, +5.17] | +3.80 [+3.17, +4.42] | +4.55 [+3.92, +5.19] |
| xp2            | +4.93 [+4.29, +5.56] | +4.91 [+4.29, +5.54] | +4.83 [+4.20, +5.46] |
| xp3            | +5.15 [+4.52, +5.78] | +4.06 [+3.44, +4.69] | +4.20 [+3.56, +4.83] |

**Table 20:** The three independently trained runs head to head. 2 of the three pairs separate from zero, which is what establishes that these are distinguishable policies rather than one policy three times.

| Pair       | Edge (pp) | 95% CI         |
|------------|-----------|----------------|
| xp1 vs xp2 | +0.20     | [−0.43, +0.83] |
| xp1 vs xp3 | +0.70     | [+0.08, +1.32] |
| xp2 vs xp3 | +1.05     | [+0.43, +1.68] |

## 11.7 RQ2: robustness to rule dialects

Every row keeps its sign, every row replicates, and the largest excursion from the default row is +0.78 pp on arb=turn, inside the registered 2.00 pp tolerance. The advantage survives 48-card Literature, all three arbitration orders, both declaration-permission changes and the full legacy dialect. Table 21 gives every row.

**Table 21:** The rule-dialect table. Both configurations play the same dialect in every row. Per-bank values are given in bank order, 7090003 then 7090001. This is the first dialect sweep of its kind in the corpus and has no training-bank counterpart.

| Dialect             | Edge (pp) | 95% CI         | Per bank      | vs default |
|---------------------|-----------|----------------|---------------|------------|
| default             | +4.73     | [+4.10, +5.36] | +4.57 / +4.88 | +0.00      |
| no-out-of-turn      | +5.25     | [+4.63, +5.87] | +4.72 / +5.78 | +0.52      |
| no-cardless-declare | +4.90     | [+4.27, +5.52] | +4.71 / +5.08 | +0.17      |
| maxasks=360         | +4.73     | [+4.10, +5.36] | +4.57 / +4.88 | +0.00      |
| arb=high            | +4.85     | [+4.21, +5.49] | +5.03 / +4.67 | +0.12      |
| arb=turn            | +5.51     | [+4.88, +6.14] | +5.26 / +5.77 | +0.78      |
| sets=8              | +4.13     | [+3.52, +4.73] | +3.77 / +4.48 | −0.60      |
| legacy              | +5.12     | [+4.50, +5.75] | +4.62 / +5.63 | +0.40      |

Two readings belong with the table. The `maxasks` row is bit-identical to the default (yes), because the 400-ask cap is never reached in this population, so it measures nothing. Consequently the legacy residual, which the protocol assigns to the forced-endgame willingness ladder for want of a flag of its own, is arithmetically legacy minus two effective components rather than three; it is −0.292 [−1.55, +0.96], an interval 8.6 times the size of the residual, so the data are equally consistent with the ladder contributing nothing.

## 11.8 RQ2: adversarial search against SESTINA v1.0

8 of eight searches produced a weight vector and all eight were evaluated. **None of the eight searches found a positive edge against SESTINA v1.0 at the tested budgets.** Every one loses, by between 2.91 and 10.70 pp; every sign replicates; and the largest upper bound anywhere is  $-2.27$ , which is 3.80 pp below the detection floor. The registered claim is satisfied. Table 22 gives every search.

**Table 22:** 8 independently constructed adversarial searches against SESTINA v1.0. A positive edge would mean the adversary beats the target. Each was fitted on a deal stream disjoint from the banks it is evaluated on. Per-bank values are given in bank order, 7090001 then 7090002.

| ID  | Class         | Objective | Edge (pp) | 95% CI             | Per bank          | Repl. |
|-----|---------------|-----------|-----------|--------------------|-------------------|-------|
| Z01 | C1 in-class   | win       | $-2.91$   | $[-3.55, -2.27]$   | $-2.76 / -3.06$   | yes   |
| Z02 | C2 extended   | win       | $-3.90$   | $[-4.52, -3.28]$   | $-3.80 / -4.00$   | yes   |
| Z03 | C1            | declerr   | $-10.70$  | $[-11.32, -10.09]$ | $-10.65 / -10.76$ | yes   |
| Z04 | C1            | events    | $-9.64$   | $[-10.27, -9.01]$  | $-9.83 / -9.46$   | yes   |
| Z05 | C1            | forced    | $-7.14$   | $[-7.76, -6.52]$   | $-7.17 / -7.11$   | yes   |
| Z06 | C2 wide sigma | win       | $-5.25$   | $[-5.88, -4.62]$   | $-4.93 / -5.57$   | yes   |
| Z07 | C1 v05 basin  | win       | $-5.21$   | $[-5.84, -4.58]$   | $-4.85 / -5.58$   | yes   |
| Z08 | C5 white-box  | win       | $-3.56$   | $[-4.19, -2.92]$   | $-3.95 / -3.17$   | yes   |

The searches control all three opposing seats, run in the synchronised correlation regime, and use a budget of 8 generations of a 12-member population at 120 deals per candidate. They maximise four different objectives, being win rate, declaration error, event count and forced-endgame incidence, and two of them depart deliberately from the template, one using a wide search variance and one seeded in the v0.5 basin, so that the set is not one search repeated. The declaration-objective arm is read against the declaration-family floor of 2.13 rather than the general floor.

This is a lower bound on exploitability and not an upper bound. It is evidence about these searches at these budgets in these classes, and §12.4 states what would be required to convert it into a statement about SESTINA v1.0.

### 11.8.1 The white-box transcript-inversion adversary

**Zeight** is the adversary class the threat model of §8 was written for: an opponent that reads the target’s own deterministic policy and reconstructs the deal posterior from the public transcript, testing directly whether common-knowledge coordination among identical deterministic teammates exposes exploitable behavioural information. It reaches  $-3.56$  pp  $[-4.19, -2.92]$ , a replicated loss.

Three limits belong with that cell, since it is the one a sceptical reader should weigh most heavily. The inverter the base fixes queries a v0.6 model of the target rather than the frozen configuration itself, a setting carried forward from the phase whose target *was* v0.6, so the likelihood is computed against a policy that does not search, leaves the urgency escalation on, and carries neither the tie-break nor the contestation coordinate; the corpus does not measure how far that model’s choice diverges from the frozen target’s, so the size of the resulting degradation is unquantified. 18 of its fitted coordinates are inert, because the white-box base derives from the v0.6 agent and reads only that policy’s 37 coordinates while the search extends its vector to 55; the effective search is therefore over 37 live coordinates plus the inverter the base fixes. And the budget is small. The result is evidence that this inverter, at this budget, does not exploit SESTINA v1.0; it is not evidence that no white-box attack can.

**Fitting performance is not evaluation performance.** The in-class arm reaches +6.67 pp of paired margin in its final generation on its own fitting stream, and +8.33 pp at its best generation, before landing at −2.91 pp on disjoint banks. Most of that difference is not the change of bank: the same run re-scores its selected vector on the fitting stream and already falls below parity there, because the generation figure is a maximum over candidates at 120 deals each rather than a measurement of the vector that was retained. Re-evaluation closes most of the gap and disjoint banks close the rest.

## 11.9 Validity controls

If these misbehave, nothing else in this section is interpretable. Table 23 is the planted-edge recovery.

**Table 23:** Planted-edge recovery. The recovered excess is the responder’s edge against the handicapped target minus its edge against the unhandicapped one. The sub-floor rung must *not* be recovered.

| Planted handicap          | Planted cost               | vs. handicapped         | vs. SESTINA             | Recovered excess                  |
|---------------------------|----------------------------|-------------------------|-------------------------|-----------------------------------|
| 0.05 ( <i>sub-floor</i> ) | +0.78<br>[+0.25, +1.32]    | −3.62<br>[−4.26, −2.99] | −4.32<br>[−4.94, −3.69] | <b>+0.70</b><br>[−0.20, +1.59]    |
| 0.08                      | +1.59<br>[+1.05, +2.13]    | −2.29<br>[−2.92, −1.67] | −4.67<br>[−5.30, −4.05] | <b>+2.38</b><br>[+1.50, +3.26]    |
| 0.11                      | +8.52<br>[+7.93, +9.12]    | +2.52<br>[+1.89, +3.16] | −6.90<br>[−7.52, −6.28] | <b>+9.42</b><br>[+8.54, +10.31]   |
| 0.15                      | +11.44<br>[+10.85, +12.03] | +7.30<br>[+6.68, +7.93] | −5.34<br>[−5.97, −4.71] | <b>+12.64</b><br>[+11.75, +13.53] |

Two of the three supra-floor rungs are recovered with a lower bound above the 1.53 pp floor. The third, the 0.08 rung, recovers +2.38 [+1.50, +3.26], above the floor on the point estimate but with an interval straddling it, so on the lower-bound reading this paper applies to its own primary claim it is not recovered. Its planted cost, +1.59 [+1.05, +2.13], straddles the floor as well, so that rung is supra-floor only on a point estimate. Recovered size tracks planted size monotonically across all four rungs.

The sub-floor rung is **not** recovered, at +0.70 [−0.20, +1.59], which is the control that has to fail and does. Two qualifications belong with it: phase 2 measured the excess at this same handicap rung at +0.25 [−0.19, +0.68], so this cycle reaches the same conclusion from a point estimate a little under half a point higher, and the upper bound here grazes the floor. The protocol had named a different phase-2 row for this comparison; Appendix E.2 records the correction.

The identity control is exact: SESTINA v1.0 against itself is 0.500000 with a zero-width interval on both banks over 12,000 games each. The responders fitted for these controls also lose to unhandicapped SESTINA v1.0 in all four rows, a second sample of §11.8’s finding from a different bank and against a handicapped target; they maximise win rate as five of the eight adversarial searches do, so this is a second sample rather than an independent method. Table 24 is the seat-isolation residual.

The gate column is zero for every configuration on both banks (0 nonzero cells), so there is no side-channel finding. The left column reproduces the known engine defect of §6.8 on holdout material and reproduces its shape: 4 cells are nonzero and every one belongs to a *searching* configuration, with the non-searching blueprint zero on both banks. The audited decision counts move between the

**Table 24:** The seat-isolation residual on holdout material, 16 cells over 19,200 audited deals. The right column is the registered zero-tolerance condition; the left column is the deployed mode.

| Configuration     | Bank    | --threads=1 |                    |
|-------------------|---------|-------------|--------------------|
|                   |         | --threads=1 | --freshagents      |
| SESTINA v1.0      | 7090001 | 0 / 813,512 | <b>0</b> / 813,512 |
| SESTINA v1.0      | 7090002 | 3 / 816,049 | <b>0</b> / 816,016 |
| v0.6 deployed     | 7090001 | 0 / 791,530 | <b>0</b> / 791,530 |
| v0.6 deployed     | 7090002 | 0 / 792,734 | <b>0</b> / 792,734 |
| <b>F-cheap</b>    | 7090001 | 1 / 792,140 | <b>0</b> / 792,140 |
| <b>F-cheap</b>    | 7090002 | 2 / 795,218 | <b>0</b> / 795,224 |
| phase-2 composite | 7090001 | 2 / 814,702 | <b>0</b> / 814,726 |
| phase-2 composite | 7090002 | 0 / 816,225 | <b>0</b> / 816,225 |

two columns in 3 of the cells, which is the measured form of the statement in §7.10 that rebuilding agents per deal changes play.

### 11.10 Computational cost

The record priced the frontier’s search at  $3.2\times$  the blueprint, a figure that describes **F-cheap**. On a like-for-like basis **F-cheap** measures at least  $4.10\times$ , and SESTINA v1.0 at least  $4.52\times$ . All three tight bounds exceed  $3.2\times$ , so the record understates the cost. The whole-panel column is lower only because expensive panel members place a large common term on both sides of the ratio, and it is the most diluted of the three. Table 25 gives the figures.

**Table 25:** Cost relative to the blueprint. Every figure is a *lower bound*, because the harness reports whole-match throughput, which carries the opponent’s cost on both sides of the ratio. The bound is tightest against the cheapest opponent. No figure carries an interval, these being wall-clock timings taken across a battery that ran over two days, and the aggregation is a ratio of medians, which differs from a median of per-bank ratios by at most 0.041 across the six median cells.

| Configuration     | vs <b>random</b> (tightest) | Far archetypes (median) | Whole panel (median) |
|-------------------|-----------------------------|-------------------------|----------------------|
| SESTINA v1.0      | $4.52\times$                | $4.72\times$            | $2.72\times$         |
| <b>F-cheap</b>    | $4.10\times$                | $4.36\times$            | $2.63\times$         |
| phase-2 composite | $4.20\times$                | $4.51\times$            | $2.69\times$         |

## 12 Discussion

### 12.1 What SESTINA v1.0 improves

SESTINA v1.0 is the strongest configuration this project has produced. Against **F-cheap**, the cheapest configuration genuinely on the v0.6 frontier and the target registered in advance, it gains 3.33 pp of win-rate edge over 48,000 games of material sealed before it existed, with the sign replicating on both banks and every advance-specified validity check satisfied. It gains +4.63 pp on the deployed v0.6 policy and +5.18 pp on v0.5.

The advantage is not an artefact of self-play coordination, which is the failure mode the cooperative-play literature exists to warn about [39, 42]. A single seat’s upgrade is worth +1.93 pp

of the +4.49 pp three-seat gain, so most of the value is individual rather than conventional; cross-play between three independently trained runs moves the result by 0.01 pp; and the advantage survives eight rule dialects with a largest excursion of +0.78 pp.

Answering RQ3, the gain is attributable primarily to one component, a half-suit contestation weight whose leave-one-out drop is +1.69 pp [+0.80, +2.57], the largest of the five. The stronger statement, that this coordinate is where the gain lives, is not supported at this power (§11.4). That component originated in a coordinate sweep during adversary generation rather than in architecture work; its value was chosen as the argmax of a measured dose response, and the mechanism it turned out to express is not the one it was designed for (§6.3). Its provenance matters for the next subsection.

## 12.2 What SESTINA v1.0 does not improve

Against the phase-2 composite, SESTINA v1.0 gains +0.15 pp [−0.29, +0.59], an interval containing zero (§11.2.1). The composite is not an external baseline; it was assembled during this cycle’s own development, from the v0.6 frontier’s search, the contestation weight and one switch. So the correct account of this cycle’s progress is that **the strength gain is real and was produced during adversary generation, and the architecture work that followed contributed no measurable strength on top of it.**

Three further results are consistent with that account and none is comfortable. The frozen configuration is 3rd of four on minimax regret over the shared panel. Its headline worst cell does not replicate in sign. And of the four quantities the registered selection check tabulates, the largest shortfall against an advance-stated training value falls on precisely the group of components the configuration was selected over.

Four candidate mechanisms were tested and rejected (§9.4), and each constrains a previously plausible direction rather than merely failing. The declaration-allocation channel, ranked second in the project’s design register, is worth zero because the mechanism it posited does not exist: the sequential joint’s first chain-rule factor is the marginal, so the two allocators agree at every recorded decision. A leaf evaluator improved 74-fold in the only statistic its consumer can use changes play by nothing, because the deviation rule that consumes it is nearly leaf-invariant, which is the abstraction pathology of [66] appearing in a new domain. Each of the three self-oriented per-decision objectives tested produced a worse policy at matched budget while moving its own proxy in the intended direction; the two channel-oriented fits lost because the ask and declaration channels trade against each other. And a learned re-ranker aimed at amortising test-time search loses outright deployed unrestricted at −1.83 pp; restricted to the exchangeable tied set, nothing it learned survives a free public-history tie-break. Inside that tied set approximately 95.5% of the search’s own selection signal is winner’s curse on its Monte Carlo draw, against approximately 84% for the search as a whole.

## 12.3 Limits of the current policy class

Answering RQ4, the natural summary is that this policy class is flat, and that summary would overstate the evidence. What the evidence supports is narrower.

**At the calibrated detection scale of this evaluation, the tested policy class appears locally flat: none of the investigated architectural mechanisms produced a replicated gain above the instrument’s effective resolution.** The resolution is 1.53 pp for a fitted responder and roughly one point for a scoreboard comparison at the budgets used here.

That is not a claim that no further gains exist, and two specific gaps are worth naming. The evaluation resolves effects of about a point and does not resolve effects of a quarter of a point, so a family of small mechanisms that individually sit below the floor and compound is entirely consistent with everything reported here. Neither this cycle nor the last tested that hypothesis, and both said so. Separately, the detection floor does not fall as evaluation games increase (§7.9), so the constraint is search power rather than sample size, and lowering it is a research problem rather than a compute problem.

The one mechanism that did convert was found by sweeping a coordinate, not by designing an architecture. That is a weak signal about where remaining value lies, but it is the signal the data provide.

## 12.4 Exploitability, and what the adversarial results do and do not show

None of the eight adversarial searches found a positive edge against SESTINA v1.0 at the tested budgets, in the tested classes, controlling all three opposing seats in the synchronised correlation regime. Every one of those qualifiers is load-carrying, and the result is a *lower bound* on exploitability rather than an upper bound.

A stronger response outside the tested classes, or the same classes at a larger budget, could give a larger figure. The white-box transcript inverter, which is the class this domain’s threat model was written for, effectively searched 37 live coordinates at 8 generations of 12, against a v0.6 model of the target rather than the target itself (§11.8.1). Its failure is evidence about that search. This is the precise reading error the v0.4 record in this corpus committed, having reported a null exploitability result as a demonstration of robustness, and naming it here is how this paper avoids repeating it.

Converting any of this into a statement about SESTINA v1.0 rather than about the searches would require a bound in the other direction, and §12.7 explains why none exists.

## 12.5 External validity

Three limitations bound how far any of these results generalise.

The opponent panel is a modelling choice. Its deception archetypes are stylised implementations of manoeuvres one person reported from live play, and they are not people. A gain against one of them is evidence about robustness to a class of behaviour, not a measurement of play against a person. The partner-substitution columns have the same limitation from the other side, since a foreign partner here is another bot.

There is no human evaluation of any kind in this corpus. The project’s standing practice is never to present a self-play configuration as the one a person would encounter, and that practice exists precisely because the substitute is not the thing.

Every number in this paper was produced by one engine, written by this project, audited by tests this project wrote. The strongest available substitute for external replication was used, in the form of material sealed before the configuration existed, digests computed by a second build, and a protocol registered in advance, and it remains a substitute. In particular, there are no independently authored six-player Fish agents to compare against, which is why §1.2 confines the paper’s comparative claim to this project’s own lineage.

## 12.6 Engineering and methodological limitations

**A known engine defect is present in every number.** Agents are constructed once per worker thread and reused across deals, and some per-agent state survives the inter-deal reset, reachable

only under the search. The carrier was identified as the rollout engine, the specific field was not, and the repair was deliberately deferred because it changes the play of every searching configuration by about one deal in 800 and applying it after freezing would defeat the purpose of freezing. §11.9 measures it on holdout material: 4 of 16 audited cells show a nonzero residual in the deployed mode, every one of them a searching configuration, and the zero-tolerance gate condition is clean.

**The soundness gate certifies a mode no scored cell uses.** Its zero-tolerance form requires one thread and agents rebuilt per deal, and rebuilding agents per deal changes play. The residual is not removed from the deployed mode.

**Cross-play cannot detect an inherited convention.** The three independently trained runs of §11.6 all begin from or near the v0.6 vector. A convention all three inherited from that common ancestor would appear on the diagonal and off it alike and would be invisible to the test. What the test rules out is a convention formed during each run’s own fit.

**Four sealed banks were digest-verified and never played.** The fitting driver derives its own per-generation deal stream from its seed rather than constructing the named bank’s deals, so three banks reserved for fitting are verified and then unused as deals; a fourth is unused because the sealed panel members were scored on the two primary banks so that minimax regret is computed over one shared pair. The material accounting should be read as three evaluation banks played plus a derived fitting stream.

**One dialect row measures nothing.** The ask-cap row is bit-identical to the default, so the legacy residual it feeds is arithmetically legacy minus two effective components rather than three, with an interval 8.6 times the size of the residual itself.

**Termination is empirical rather than proved.** No configuration here reaches the action cap in any scored cell but one, in 12,000 games, and the stall detector’s separation between ordinary play and deadlock is a measurement over 800 mirror games. What is not proved is that the stall rule together with the stopping rule terminates on every reachable position.

## 12.7 Future work

Two claims a future release might wish to make, that a Fish agent is near-optimal or that it is the strongest known, are unsupported by anything here. Both are reachable in principle, and the evidence each requires is specific enough to enumerate.

**An upper bound on exploitability.** Every exploitability figure in this corpus is a lower bound produced by a search. A near-optimality claim requires a bound in the other direction, showing that no adversary in a stated class can exceed some  $\varepsilon$ . None exists here for any class, and producing one even for a restricted class would convert every exploitability sentence in this corpus from a statement about searches into a statement about the agent.

**A team analogue of local best response, and the rollout policy it needs.** The standard cheap exploitability probe rests on a passive continuation [38]. Fish has none, since a player must ask or declare (§2.3). Until a defensible rollout policy for that role is named and calibrated the probe cannot be ported, and the literature’s own caution that choosing a value function is the difficult part applies directly [24].



**A genuinely correlated adversary.** The threat model names three correlation regimes and makes the ex-ante correlated one the principal case. What has been built and measured is the synchronised regime, plus one correlated fit reaching +0.70 pp against a class floor of 2.31, which is nothing. Whether such a responder is buildable for a game of this size is open, and the honest position is that the principal regime has fallen back to a weaker one.

**Whether small effects compound.** This is the one hypothesis both this cycle and the previous one explicitly declined to test. Resolving a point on the scoreboard requires roughly 9,604 games per cell, and a quarter of a point sixteen times that, while the per-decision estimator of §9.4.3 resolves the same one-point effect in 284 to 560. That estimator is confirmed and unused, because fitting against it fails. Using it as an estimator while continuing to fit against the scoreboard is the obvious untried experiment.

**External opponents and human play.** A comparative claim beyond this lineage requires at least one independently authored opponent and, for any claim about play against people, a duplicate-style human battery with enough deals to resolve a point. Neither exists, and the six-seat duplicate design for the latter has not been settled.

**Repairing the cross-deal residual.** The defect of §12.6 should be repaired and everything it moves re-measured, before rather than after any subsequent configuration is frozen.

## 13 Conclusion

SESTINA v1.0 is the strongest agent this project has produced. Against **F-cheap**, the cheapest configuration genuinely on the v0.6 frontier and the comparison target registered before any holdout deal was played, it gains 3.33 pp of team win-rate edge [+2.88, +3.78] over 48,000 games of sealed material, with the sign replicating on both banks and every advance-specified validity condition satisfied.

No coordination or robustness test in this paper overturns that advantage, though the picture they give is mixed. A single seat’s upgrade purchases +1.93 of the +4.49 pp three-seat gain, so the value is largely individual rather than conventional; cross-play between three independently trained runs moves the result by 0.01 pp; the advantage persists across eight rule dialects; and none of eight independently constructed adversarial searches, including a white-box adversary that inverts the target’s transcript against a v0.6 model of its deterministic policy, found a positive edge at the tested budgets. Two of those tests are not clean: the partner row that decides the coordination claim is  $-0.19$  [ $-1.04, +0.67$ ], above the registered  $-1.00$  collapse threshold on the point estimate but not resolved against it (§11.5); and over the shared panel SESTINA v1.0 is 3rd of four on minimax regret with a worst cell that does not replicate in sign (§11.3).

Two further findings qualify that picture and are the reason this paper is worth more than its headline. The first is that SESTINA v1.0 does not measurably outperform a composite configuration assembled during its own development, at +0.15 pp [ $-0.29, +0.59$ ]. The improvement in this cycle is therefore real, but it was produced during adversary generation by a coordinate sweep whose value was the argmax of a dose response, and the architecture work that followed added no measurable strength on top of it. The second is that four plausible candidate mechanisms were experimentally rejected, each with a diagnosis rather than merely a null: the declaration-allocation channel because the mechanism it posited does not exist, a fitted leaf evaluator because the rule consuming it is nearly leaf-invariant, per-decision objectives because the ask and declaration channels

trade against each other, and a learned re-ranker because it loses outright when deployed unrestricted and, restricted to the exchangeable tied set, is indistinguishable from a free public-history randomisation.

Taken together, progress in this policy class is now limited by effects smaller than the evaluation and adversary-search machinery can reliably detect. At the calibrated detection scale of this study the class appears locally flat, in the sense that no investigated architectural mechanism produced a replicated gain above the instrument’s resolution. That is not a demonstration that no further gains exist. The evaluation resolves effects of about a point and not of a quarter of a point, so a family of small mechanisms that compound remains consistent with everything reported here, and testing that possibility is the clearest next step.

Finally, the limits of what this study establishes are worth restating, because they are what the evidence permits. SESTINA v1.0 is the strongest configuration in this project’s lineage. It is not shown to be near-optimal, it is not shown to be unexploitable, and it is not shown to be the strongest Fish-playing agent in existence. The comparison class contains only agents this project wrote, no human evaluation exists, and every exploitability figure here is a lower bound produced by a search rather than a bound on what a search could find. §12.7 sets out the evidence each of those claims would require. Producing a stronger Fish agent turned out to require not only better play but a substantially better method for deciding whether apparent improvements were real, and the second of those is the part most likely to transfer.

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## A The rules dialect, clause by clause

Six of the eight contested rule choices of §2.2 are engine switches, so their effect is measurable rather than assumed; the misdeclaration award is hardcoded, and the forced-endgame ladder has a threshold flag but not one the scored harness exposes, so within a scored match it moves only inside the `--legacy` bundle. This appendix lists each choice against the switch that changes it, so that a reader can re-implement the dialect or vary it.

| Clause                  | Switch                                            | Setting studied                                                                               |
|-------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Deck and half-suits     | <code>--sets=8</code>                             | 54 cards in nine half-suits; <code>--sets=8</code> plays the 48-card, eight-half-suit variant |
| Out-of-turn declaration | <code>--no-out-of-turn</code>                     | permitted; the switch forbids it                                                              |
| Cardless declaration    | <code>--no-cardless-declare</code>                | permitted; the switch forbids it                                                              |
| Simultaneous declarers  | <code>--arb=high</code> , <code>--arb=turn</code> | lowest seat index first; the switches arbitrate by stated confidence and by turn order        |
| Misdeclaration          | —                                                 | the half-suit is awarded to the opposing team                                                 |
| Forced endgame          | —                                                 | eight-level willingness ladder when one whole team is cardless                                |
| Action cap              | <code>--maxasks=N</code>                          | 400 asks; residue adjudicated by physical majority                                            |
| Legacy bundle           | <code>--legacy</code>                             | the full v0.3-era dialect, bundling several of the above                                      |

§11.7 measures the paper’s headline comparison over 8 dialect rows: the default, and seven variations drawn from six of these clauses. Every row keeps its sign and every row replicates; the largest excursion from the default row is +0.78 pp, on `arb=turn`.

**Two clauses on which an earlier report in this corpus is wrong, corrected here.**

1. **An undeclared half-suit does not score nothing.** An earlier report justified an unconditional forcing rule on the premise that an unclaimed half-suit is worthless. Under the dialect actually evaluated, residue at the action cap is awarded by physical majority, and on a wrong post-horizon declaration the declaring team holds a mean 4.08 of the six cards. The rule converts a probable award into a certain concession.
2. **Restarting the forced-endgame ladder sharpens nothing.** The argument is that restarting the willingness ladder after each declaration lets early, safer declarations resolve cards and so sharpen later ones. It is sound and, in the play it was measured on, inoperative: on every occasion a team went cardless with live half-suits — 566 such records at seed 31 in v0.4 mirror self-play, half of them duplicates, because at each deal shift the two team orientations play identical policies and so record every event twice — exactly 1 half-suit was live, so there was no ordering to exploit. That is a strong tendency and not a universal: a second sample at seed 1234567 finds 60 *distinct* forced declarations, all but one of them with a single live half-suit. It is also policy-dependent: the source records that a policy made more patient about declaring would begin to produce multi-half-suit forced endgames, at which point the ordering would start to matter. SESTINA v1.0 is a change in that direction, since it disables the urgency escalation entirely (§6.5), and no measurement of live-half-suit multiplicity at forced-endgame entry exists for it.



## B Configuration grammar and parameters

### The grammar

A configuration is named by a single string, and every policy in this paper, every arm, every panel member, every fitted adversary, is one of these strings. The grammar is

```
base [ :key=value ( ,key=value )* ]
```

where `base` names a policy family (`v03 ... v07`, a scripted archetype such as `withholder`, or an instrumented variant such as `v07i` for the white-box inverter), and each `key=value` sets one option on it. Two properties of the grammar matter for the evaluation:

- **It is total.** Any configuration this paper measures can be written as one string and reconstructed from it, which is what makes an ablation lattice mechanical rather than a set of hand-edited builds.
- **An explicit key beats a bulk vector.** A spec may carry `allparams=w1|w2|...`, the entire parameter vector; explicit keys written alongside it are re-applied afterwards. Before this was fixed, a sentinel such as `pool=-1` was silently clamped away by the bulk vector, and three independently fitted configurations carried machinery their spec said was switched off.

### The frozen configuration

```
v07:r12=25,rtie=1,pool=-1,oppfloor=-1,force=1000000,askfloor=-1,stall=12,s1=1,
det=12,cand=4,kappa=2.5,rbelief=indep,depth=12,maxq=26
```

| Key                   | Value              | What it controls                                                                |
|-----------------------|--------------------|---------------------------------------------------------------------------------|
| <code>r12</code>      | 25                 | half-suit contestation weight ( <code>oppCertDonate</code> )                    |
| <code>rtie</code>     | 1                  | tie-break by a hash of the public event stream, not sort order                  |
| <code>pool</code>     | -1                 | urgency pooling threshold; -1 disables                                          |
| <code>oppfloor</code> | -1                 | urgency opponent-ownership floor; -1 disables                                   |
| <code>force</code>    | 1000000            | urgency forcing horizon in events; 10 <sup>6</sup> disables                     |
| <code>askfloor</code> | -1                 | urgency ask floor; -1 disables                                                  |
| <code>stall</code>    | 12                 | escalate after $K$ public events with no change in this seat's certificate hash |
| <code>s1</code>       | 1                  | enable the endgame-truncated determinized search                                |
| <code>det</code>      | 12                 | determinizations sampled per searched decision                                  |
| <code>cand</code>     | 4                  | candidate actions carried into the search                                       |
| <code>kappa</code>    | 2.5                | shrinkage of the paired lower-confidence-bound deviation rule                   |
| <code>rbelief</code>  | <code>indep</code> | belief model used to reconstruct continuation players                           |
| <code>depth</code>    | 12                 | search depth limit in plies                                                     |
| <code>maxq</code>     | 26                 | search runs only when at most this many cards remain unresolved                 |

Three of these are switches that cannot be expressed as coordinates of the parameter vector at all (`askfloor`, `pool`, `oppfloor`), which is why the freeze artifact records the option map separately from the vector and why the round-trip check of §6 has to execute rather than compare strings.

## The parameter vector

| Block        | Coordinates |
|--------------|-------------|
| nfeat        | 20          |
| v05knobs     | 14          |
| v06askTerms  | 3           |
| v07responder | 18          |
| total        | 55          |

the flat `allparams` vector: NFEAT ask weights, the fourteen v0.5 knobs, the three v0.6 ask terms, then the eighteen v0.7 responder coordinates (`v07_responder.hpp r7Name`)

The first 37 coordinates are inherited unchanged from the previous release’s fit. The freeze artifact records its provenance as

```
fitC.jsonl [final weights record] sha256:be4a4c31d9153208 gens=14 obj=minimaxregret
paired=1 panel=v05+v03+withholder+fe...
```

The remaining coordinates are the extended responder block, and they are all zero but one. That is the identity condition: with every responder coordinate at zero the extended class reproduces v0.6 bit for bit, which is the certificate §6 relies on. No gradient or population fit was run in this cycle. The single non-zero coordinate, `r12`, had its value chosen as the argmax of a measured dose response (§6.1, §6.3).

## Coordinates known to be inert

Not every fitted coordinate does anything, and a paper that presented all 55 as tuned quantities would be overstating the search. Known inert coordinates in this family:

- two v0.5 knobs unreachable under the deployed declaration branch, so setting them produces a bit-identical transcript;
- three v0.5 ask terms belonging to a re-scoring pass that the deployed configuration does not run;
- `m2`, whose leave-one-out drop is exactly zero and which was dropped from the freeze for that reason (§11.4);
- `stall`, which is armed and never fires: its mirror digest is byte-identical to the same configuration without it (§6.6);
- and, in the white-box adversary class specifically, 18 of the 55 fitted coordinates, because that base reads only the 37-coordinate v0.6 vector (§11.8.1).

## C The preregistered thresholds, verbatim

Reproduced from `docs/v07/PREREGISTRATION.md`, committed before any holdout bank had been played and before the sealed adversary half had been decoded. The verdicts are §11 and §11.2.

### The primary claim

SESTINA v1.0 beats **F-cheap** on both primary holdout banks, pooled over 48,000 games.

| Verdict               | Condition                                                                            |
|-----------------------|--------------------------------------------------------------------------------------|
| CERTIFIED advancement | pooled lower bound above 1.53, sign replicates on both banks, gate passes            |
| measured advancement  | pooled lower bound above 0 but under 1.53, sign replicates, gate passes              |
| not an advancement    | the pooled interval contains zero, or the sign does not replicate, or the gate fails |

The verdict labels above are the protocol’s own. This paper does not use the word *certified*, which suggests verification by a third party, and reports the top outcome instead as what was actually satisfied (§10.4). Result: the comparison **satisfies the registered primary criterion**.

## The secondary claims

| #  | Claim                              | Passes if                                                                                       | Result |
|----|------------------------------------|-------------------------------------------------------------------------------------------------|--------|
| S1 | no collapse under partner change   | at least five of the eight changed-partner deltas positive, and the minimum not below $-1.00$   | PASS   |
| S2 | not a private convention           | the off-diagonal is within 1.50 points of the diagonal                                          | PASS   |
| S3 | no fresh adversary exploits it     | every arm’s upper bound below its floor                                                         | PASS   |
| S4 | the gain is attributable           | components sub-additive, no single drop exceeding the whole                                     | PASS   |
| S5 | it survives the rule dialect       | every row keeps its sign, none more than 2.00 pp below default                                  | PASS   |
| S6 | the worst case is not catastrophic | SESTINA v1.0’s worst panel cell no worse than the deployed v0.6 policy’s                        | PASS   |
| S7 | the instrument is intact           | the planted edges recover, the sub-floor rung does not, the cheats fail exactly their own tests | PASS   |

The protocol’s “eight changed-partner rows” counts the self-play row among them; §11.5 reports 6 of the 7 genuinely changed rows. Both readings pass, and the phase-5 recording carries the ambiguity as deviation D9.

A failure of S1, S2 or S6 would have been a substantive failure, and the paper would report SESTINA v1.0 as brittle in that named way. A failure of S7 would have been procedural and the evaluation would have stopped.

## What would mean v0.7 is not an advancement

The seven conditions are answered in §11.2.4. Two are met or unresolved. The protocol’s own closing provision is quoted in full, because this paper relies on it:

“And the honest all-negative path is a legitimate deliverable. If phase 5 returns ‘no advancement over the frontier’, the v0.7 report is a barrier report: the four ledger entries phase 3 closed with measurements, the two corrections to the corpus’s own record, the mechanical side-channel gate with its three calibrated positive controls, and the S6 residual this document commits. That is a real contribution and the report should not apologise for it.”

## D The full panel

Every cell of the shared panel, four arms against 31 members, pooled over both primary banks. It is printed in full so that a reader can take the worst cell themselves rather than take §11.3’s word for which it is, and so that the minimax regret of Table 13 can be recomputed by hand.

Class **near** is 6,000 deals a bank, **far** 1,500, **expensive** 1,200. A positive number means the arm beats that member.

| Panel member         | Class     | SESTINA v1.0 | v0.6   | F-cheap | Composite |
|----------------------|-----------|--------------|--------|---------|-----------|
| F-fast               | near      | +4.55        | +0.00  | +1.83   | +4.17     |
| F-cheap              | near      | +3.38        | -1.83  | +0.00   | +2.37     |
| v0.5                 | near      | +5.18        | +0.69  | +1.84   | +4.45     |
| v0.4                 | near      | +4.66        | +0.99  | +2.09   | +3.86     |
| feint                | near      | +6.07        | +3.32  | +4.61   | +4.93     |
| SEALED:R-v04         | near      | +4.66        | +0.99  | +2.09   | +3.86     |
| SEALED:S-archetype-0 | near      | +6.07        | +3.32  | +4.62   | +4.93     |
| SEALED:S-ask-1       | near      | +4.62        | -0.53  | +1.01   | +3.29     |
| SEALED:S-ask-3       | near      | +4.08        | -0.22  | +1.55   | +3.85     |
| SEALED:S-decl-0      | near      | +4.06        | -0.72  | +1.10   | +3.68     |
| SEALED:S-invert-0    | near      | +3.93        | -1.74  | +0.90   | +1.90     |
| SEALED:S-reference-0 | near      | +4.55        | +0.00  | +1.83   | +4.17     |
| SEALED:S-search-1    | near      | +3.43        | -1.81  | +0.07   | +2.97     |
| SEALED:X01           | near      | +3.38        | -0.75  | +1.64   | +3.35     |
| SEALED:X01xC3f       | near      | +1.43        | -4.24  | -1.25   | +1.23     |
| SEALED:X02           | near      | +3.60        | -1.37  | +1.17   | +3.04     |
| SEALED:X13           | near      | +4.84        | -0.30  | +1.82   | +4.12     |
| SEALED:X18           | near      | +3.37        | -0.46  | +1.75   | +3.22     |
| SEALED:X20           | near      | +13.36       | +11.62 | +13.54  | +13.40    |
| v0.3                 | far       | +28.72       | +25.47 | +24.72  | +28.40    |
| v0.2                 | far       | +34.32       | +31.68 | +32.67  | +33.17    |
| lockout              | far       | +30.78       | +28.65 | +28.97  | +29.07    |
| detective            | far       | +29.65       | +28.08 | +27.98  | +28.32    |
| hunter               | far       | +46.93       | +48.58 | +48.65  | +47.10    |
| diversifier          | far       | +43.40       | +44.90 | +45.18  | +43.32    |
| bluffer              | far       | +49.87       | +49.98 | +49.95  | +49.88    |
| silent               | far       | +32.75       | +34.23 | +34.05  | +30.92    |
| withholder           | far       | +25.35       | +29.88 | +29.55  | +25.80    |
| random               | far       | +50.00       | +50.00 | +50.00  | +50.00    |
| F-mid                | expensive | +2.15        | -3.15  | -0.10   | +2.00     |
| phase-2 composite    | expensive | -0.04        | -3.69  | -2.08   | +0.00     |

## E Corrections to the earlier record

Earlier work in this lineage contains measurement and implementation errors that later cycles found. The project’s convention is that these are stated as corrections rather than silently patched into the artifacts they correct, and the itemised register is placed here so that it does not compete with the subject of the paper. §4 summarises the consequences that bear on SESTINA v1.0: earlier evaluation lacked the power to resolve the effects it reported on, one exploitability interpretation was substantially too strong, an exact-belief implementation was unsound in multi-agent configurations, and the cost of the frontier’s search was mischaracterised by two orders of magnitude. Those discoveries are why the evaluation framework of §7 exists in the form it does.

## E.1 Corrections to the earlier lineage

**The cost of the frontier’s search, wrong in both directions.** The v0.6 paper recorded the searching configuration at “three orders of magnitude” the blueprint, mixing an all-threads timing with a single-thread one; phase 1 of this cycle re-timed it on one basis at  $242\times$ . Both figures describe the *unrestricted* configuration, which is not the point of the frontier anything was measured at. The endgame-restricted operating point was re-measured at  $\sim 3.2\times$  during phase 3, and that correction was carried into the protocol. **This report corrects the correction:** measured on the shared panel at a fixed thread count, **F-cheap**, the configuration the  $3.2\times$  describes, costs at least  $4.10\times$  the blueprint, and SESTINA v1.0 at least  $4.52\times$  (§11.10). Every figure in that table is a lower bound and none carries an interval. The record has now been wrong in both directions and the reason is the same both times, a ratio quoted without the basis it was computed on.

**The corpus’s exploitability figure is a lower bound, and has been read as more.** The v0.4 findings document reports that a policy fitted specifically against v0.4 fails to reach parity with it; the same study’s own table records the response reaching 51.19% with an interval spanning parity. The v0.6 study corrected this. It is repeated here because this cycle’s own strongest result, §11.8, eight searches, all losing, is a number of exactly the same kind, and the correction is what stops this paper making the same claim.

**The declaration-allocation channel is worth zero, not two percentage points.** The register ranked it second by priority, at 0.48. §9.4.1 closes it at exactly zero over  $\sim 210,000$  recorded declarations, shows its proposed fix is worse than the object it would replace, and shows most of the residual error class is an information limit rather than a defect.

**The forced-endgame entry’s own number was wrong.** Re-derived from this cycle’s capture, closing the entire accuracy gap is worth 0.030 pp, the register’s figure corrected *upward* by a factor of two, and still a fiftieth of the detection floor. The entry is closed either way.

**An annotation in this project’s own bibliography overstated its source.** The vetted entry for the local-best-response result annotated it as showing “every ACPC 2016 entrant” exploitable above the stated bound. The source says *each evaluated bot*, and states that the competition’s winning submission was not evaluated; it also describes the search as looking one action ahead, where the annotation said two. Both are corrected in §3.3 and in the bibliography entry, whose header records the edit. This is the same widening this paper’s own adversarial passes kept finding in its prose, committed in a place nobody re-reads.

**The detection floor does not buy down with evaluation games.** Phase 1 measured 1.68 and predicted the floor would fall as the inverse square root of evaluation games. At exactly four times the games it is 1.53, and the sub-floor rung is still undetected because the excess resolves *to* zero rather than because the interval is wide. Below roughly a point and a half the responder is empty, not unresolved. This report reproduces that on holdout material (§11.9).

## E.2 Corrections this cycle makes to its own record

**An artifact was excised from version-control history after the evaluation.** A per-decision search capture from the development phase, 139,267,276 bytes, exceeded the repository host’s file-size limit and was removed from git history so the work could be published. The file is retained on disk with its SHA-256 recorded in a sidecar (bb655aac7cd413d2...) and is regenerable from the capture harness; no measurement in this paper depends on it. The rewrite changed every commit hash from phase 3 onward, so hashes recorded inside the phase-5 artifacts no longer resolve. Appendix G gives the mapping and confirms the seal commit is unaffected.

**A freeze-verification step could not fire, and its silence was not a pass.** The freeze checker reads its source-drift baseline out of the freeze artifact’s own provenance field, and that artifact was rewritten twice after the freeze commit, each rewrite refreshing the baseline to the then-current tree. The check therefore compared the tree against a baseline taken from the tree, and the sources the protocol predicted would be listed as changed no longer differed from it. Measured against the actual freeze commit, 9 of the 78 engine sources differ, `arena.hpp`, `factory.hpp`, `fish.hpp`, `game.hpp`, `human.hpp`, `main.cpp`, `table.hpp`, `tuner.hpp`, `v07_side.hpp`, of which 4 arrived in the commits that landed while the battery ran and 5 already differed before it began. **What is not weakened is the policy identity:** the round-trip assertions and the mirror digest are computed by *running* the engine, not by comparing file hashes, and all of them round-trip. Because the hash-based check was vacuous it was replaced by an executed one, the engine was rebuilt from the tree as it stood at the end of the battery and the mirror digest recomputed with the fresh binary. It is `5f81f440fc9c272a87e87c05fecc7b74`, identical to the frozen artifact and to the binary that played every cell.

**A scored cell is not bit-reproducible for a searching arm.** Before any holdout material was spent, one training cell was checked to be bit-identical over 5 runs across three thread counts, and it was (yes). **That check was too small.** The panel contains three policies twice, so twenty-four pairs of byte-identical commands were run twice each; one pair disagreed, by one game in 12,000. That is the cross-deal agent residue of §11.9 reaching the *scored* mode, which four hundred deals was not large enough to expose. The magnitude is roughly  $76\times$  below a cell half-width and no verdict in this paper turns on it, but the earlier claim should be read as “reproducible at four hundred deals”, not “reproducible”.

**The scored-cell count was not the count the artifacts support.** The phase-5 recording stated 409 scored match cells in three places; the figure was hand-typed in the reduction script rather than computed. Counting rows that carry a match object across every scored battery gives **428**. Nothing turned on it, it was the denominator of an observation about a single action-limit hit, but a corpus whose standing rule is that no number is hand-typed should not carry a hand-typed total. The generator for this paper computes it, and the phase-5 recording has since been corrected to agree.

**The partner-regime table was published without its own intervals.** The phase-5 results document printed the partner deltas as bare point estimates with a replication flag. Every underlying cell carries a deal-clustered interval, and the delta arithmetic is fixed by the protocol, so the intervals were computable and had simply been compressed out of a markdown table. Table 16 carries them, and the phase-5 recording has since been corrected to carry them too, at which point the row that decides S1 turned out to have an interval crossing its own threshold (§11.5). No verdict changes; a reader’s ability to judge the verdict does, which is the whole argument for printing intervals.

**The cross-play gap was printed under the reversed sign convention.** The phase-5 document reports the gap as  $-0.01$ , which is diagonal minus off-diagonal. The claim it settles asks whether the *off*-diagonal collapses, so the informative direction is off-diagonal minus diagonal, which is  $+0.01$ . *Neither sign is resolved:* the magnitude is 0.01 pp against a per-cell half-width of 0.63 and a 1.50-point threshold, so no verdict moves and this paper reads no lift out of it either. What the correction is worth is only this: the sign as printed *implies* a small collapse, and the data support no direction at all.

**The vetted bibliography carried eight metadata errors, four of them author lists.** Every one of the seventy-one references this paper cites has now been checked against its source rather



than against the earlier cycle’s bibliography, which is where all of them were inherited from. Four author lists were wrong: one entry carried the adjacent entry’s five names where the paper it cites has four, one printed nine names from a later paper by overlapping authors where the cited preprint has seven, and one added an author the cited work does not have. The others were a repository start year a year early, a page range, the title of a containing work, and a web article’s site printed as its author. All eight are corrected in this document’s reference list and none of them touches a claim in the text. The lesson is the one §E.1 already draws about an annotation in the same file: a bibliography that has been vetted once and copied forward is not the same thing as a bibliography that has been checked.

**The legacy residual’s interval was computed over an inert cell.** The phase-5 document forms the residual’s half-width over five dialect cells with unit coefficients, one of which is the `--maxasks=360` row. That row is bit-identical to the default — the same win rate and the same interval on both banks — so it carries no variance, and its own document says so two lines later. Excluding it, and giving the default row the coefficient the two-component residual actually gives it, the interval is  $[-1.55, +0.96]$  rather than the recorded  $[-1.95, +1.37]$ . This paper prints the narrower of the two. The two documents also express the width differently, this one as the full interval over the residual and the recording as the half-width over it, so their multiples are not comparable as printed and are omitted here. No verdict moves: S5 is decided by the largest excursion, and the residual is consistent with zero under either. Neither figure is a true 95% interval, because all the cells share banks and deals and the ask-cap identity proves the correlation is large.

**The sub-floor control’s registered comparison named the wrong handicap rung.** The preregistration introduces B9.3’s sub-floor rung as “`hstr=0.05`, the rung phase 2 measured at  $+0.88$ ” and directs the phase-5 excess to be read against phase 2’s  $-0.01$   $[-0.45, +0.42]$ . Phase 2 measured `hstr=0.05` at a planted cost of  $+0.67$  and an excess of  $+0.25$   $[-0.19, +0.68]$ ;  $+0.88$  and  $-0.01$  are its `hstr=0.08` row, and phase 1 pins the same split independently. The phase-5 recording inherited the pairing. §11.9 quotes the rung-matched figure. The verdict does not move — the control is whether the planted edge is recovered, and it is not on either pairing — but the margin it is compared against does, from seven tenths of a point to a little under half a point.

**The live-ask gate’s binding rate was the incumbent’s, and its sample size a third cell’s.** The phase-2 mechanism characterisation captures the *target* arm, so every rate in its table is measured on v0.6. Its rate column comes from a 4,000-deal mirror cell and its `n` column from the 3,000-deal treatment cell, so the pair 0.00994 over 260,987 decisions is not any one cell. The quantity also does not carry over: the gate acts on the argmax of an ask score that SESTINA v1.0 extends with a coordinate v0.6 does not have, so the rate at which it displaces that argmax is a different number. §6.1 now quotes a capture of the frozen configuration from its own seats on the same diagnostic bank, which binds at 1.4% of 269,423 ask decisions, half a point above the incumbent’s rate.

**A counter-example was quoted at twice its distinct size.** The second sample that refutes the one-live-half-suit universal (§A) is reported in its source as 118 of 120 forced declarations. That source’s own preceding section establishes that a mirror run at `--rotations=6` plays identical policies in both team orientations and so records every declaration twice, and re-runs this very seed at `--rotations=3` for 60. The distinct evidence is one occasion in 60, not two in 120. The rate is unchanged; the sample size is halved. The same doubling applies to the earlier run quoted in the same sentence, which is why it is described there as records rather than as occasions.

**A cost to an adversary was printed with no cell behind it.** The phase-2 document, closing the forced-endgame induction route, states that the construction which raises the *target*’s incidence

“was measured at  $-17$  points”. No cell in the results tree produces it: no recorded configuration string describes that construction, and the only measurement near  $-17$  in the cycle is an unrelated belief ablation. What was measured is the other construction, a team that never declares voluntarily, which raises its own incidence and not its opponent’s, at 6 to 8 pp on two independent cells. §9.4.5 now says that, and says that the route is closed for want of a mechanism rather than by a price.

**Two figures were attributed to the wrong document.** The  $242\times$  cost of the unrestricted search is phase 1 of this cycle’s own timing, produced expressly to replace the v0.6 paper’s mixed-basis “three orders of magnitude”; the protocol presented it as an inherited record that this cycle corrected. And the leaf evaluator’s 74-fold gain in between-candidate  $R^2$  is an in-sample gain, described in its source as replicating out of sample; out of sample the same pair is 0.00664 against 0.08424, so the direction replicates and the multiple does not. Neither error changes a verdict: the cost figure is corrected again by §11.10 in any case, and the leaf’s conversion failure is what kills it.

### E.3 A note on comparability

Two things about this cycle’s evidence standard are not the corpus’s, and a reader comparing across versions should have them. Every claim here is measured on material sealed before the configuration existed, under a protocol committed before any of it was played, with a stated failure condition, which no previous study in this corpus did. And every claim is replicated across two disjoint banks with the sign rule of §7.3 applied, which is why §11.2.2 reports a non-replication that a pooled-interval convention would have reported as a pass.

## F Deviations, additions and corrections

The protocol requires a deviation for anything the evaluation did that it does not specify, recorded next to the affected result rather than in a footnote. There are 27. This register is read out of `docs/v07/FINAL-RESULTS.md` by the generator rather than retyped, so it cannot drift from the results document it reports; long entries are elided with an ellipsis and the source document carries them in full.

| ID        | Kind        | Deviation, addition or correction                                                                                                                                                                                                                                                                                                                                                                |
|-----------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D1</b> | Added Cell  | B5 REF-FROZEN. The eleven cells B5 names cannot yield a leave-one-out DROP: the drop is FROZEN’s edge minus the variant’s, and FROZEN’s own edge on the lattice banks is not one of the eleven. A twelfth cell was added at the lattice size on both lattice banks. B5 therefore cost 288,000 games rather than the preregistered 264,000.                                                       |
| <b>D2</b> | Added Cells | B9 planted cost and responder-vs-unhandicapped. B9.2 requires that “the recovered size tracks the planted size”, which needs the planted size; B9.3 speaks of the planted excess “resolving TO zero”, which is the responder’s edge against the handicapped target MINUS its edge against the unhandicapped one. Four rungs x 2 banks x 6,000 deals were added for each, 192,000 games in total. |

*continued overleaf*

| ID        | Kind                           | Deviation, addition or correction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D3</b> | Unspecified Bank               | B8. Section 2.1 assigns 7090003 the role “the ablation lattice and the dialect table” but names no replicate for the dialect rows. 7090001 was used, by symmetry with B5, which the protocol does specify as 7090003 primary and 7090001 replicate.                                                                                                                                                                                                                                                                                 |
| <b>D4</b> | Unspecified Bank               | B3 sealed rows. Section 2.1 gives 7091001 the role “evaluation bank for the sealed adversary half”, but section 4/B3 puts every panel member on the same two banks as every other arm, and section 3 makes 7090001/7090002 the primary and replicate for every claim. The sealed rows were run on 7090001/7090002 with the rest of the panel, so the whole panel is scored on one shared pair of banks, as minimax regret over a shared panel requires. 7091001 was...                                                              |
| <b>D5</b> | Protocol Note                  | what <code>tune --seed=&lt;bank&gt; --shard=s/4</code> actually plays. Section 4/B4 describes the eight searches as drawing “480 deal indices of the bank”. They do not. <code>tune</code> uses <code>--seed</code> only as the CEM root: it derives a fresh per-generation seed ( <code>tuner.hpp:310</code> ) and from that a per-opponent match seed ( <code>tuner.hpp:258</code> ), so the deals are a derived stream and the named bank’s own deals are never constructed. Banks 7090004, 7090005 and 7091002 are therefore... |
| <b>D6</b> | Protocol Note                  | the seal is not enforced on every command. Section 2.1(b) records that <code>pathology</code> , <code>v7bits</code> and <code>v6probe</code> bypass <code>seedUsable</code> . <code>tune</code> and <code>ablate</code> also bypass it, for the same reason as D5: the seed <code>runMatch</code> sees is derived, not the registered one. Phase 5 set <code>FISH_UNSEAL_PHASE=5</code> once and used the seven registered seeds and no others.                                                                                     |
| <b>D7</b> | Observation, Superseded By D19 | B0.3 printed no changed-source NOTE. The protocol expected three engine sources to be listed as changed since the freeze ( <code>arena.hpp</code> , <code>main.cpp</code> , <code>v07_side.hpp</code> ). Zero differ from the baseline the check actually uses. D19 explains why that baseline is not the freeze’s and gives the correct figure – nine of 78 sources differ from the freeze commit. This entry is left standing rather than deleted, because the first reading of B0.3 was reported before it was...                |
| <b>D8</b> | Added Check                    | reproducibility, before the material was spent. <code>runMatch</code> schedules deals by work-stealing, so the deal-to-thread assignment is not fixed run to run even at a fixed thread count, and the frozen configuration carries the cross-deal agent residue of section 5.3. A scored cell was checked to be bit-identical over three runs at 13 threads and across 1, 2 and 13 threads, on training material, before any holdout bank was played.                                                                              |
| <b>D9</b> | Reading                        | S1’s row count. Section 5.2 says “five of the eight changed-partner rows”; B6 has seven changed settings plus self-play, and the protocol’s own worked table counts eight including self-play. Both counts are printed and the verdict states which reading it used.                                                                                                                                                                                                                                                                |

*continued overleaf*

| ID         | Kind             | Deviation, addition or correction                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D10</b> | Reading          | S3's pass and fail conditions are not complements. "Every arm's upper bound below 1.53" and "any arm clears 1.53 with a replicated sign" leave a band in which neither holds. Both are computed and reported; if the band is occupied the verdict is reported as UNDETERMINED rather than resolved by choosing a reading.                                                                                                                                          |
| <b>D11</b> | Reading          | the 2.13 declaration-family floor. Section 3 correction 4 applies it to "B4's Z03, and the declaration-accuracy column of B2". Z03 is read against 2.13. B2 has no declaration-accuracy column with an interval – <code>match --json</code> emits declaration accuracy as a bare rate – so the accuracy is printed as a diagnostic and PHASE 5 MAKES NO CLAIM ABOUT THE DECLARATION CHANNEL, which is what discharges the rule.                                    |
| <b>D12</b> | Arithmetic       | S4's naive sum. The freeze carries five components; <code>m2=0</code> was dropped from it. The sum S4 tests is over those five. The six-term sum including A-m2 is printed beside it and labelled, because B5 preregisters the A-m2 cell even though the freeze does not carry the key.                                                                                                                                                                            |
| <b>D13</b> | Note             | every delta in this document is two independent intervals combined in quadrature, not a paired difference. The harness gives a paired delta only within one <code>ablate</code> invocation over a shared panel, and no cell of this battery is of that form. Section 5.2 fixes this arithmetic and calls it conservative.                                                                                                                                          |
| <b>D14</b> | Note             | <code>ablate</code> was not used anywhere. It derives its own per-opponent seeds by <code>mixSeed(seed, i*7919+3)</code> ( <code>main.cpp:589</code> ), so <code>ablate --seed=7090003</code> would not play bank 7090003. Every lattice cell is a <code>match</code> cell on the named bank instead.                                                                                                                                                              |
| <b>D15</b> | Note             | the panel of 31 NAMED members contains three policies twice: R-v04 is v04, S-archetype-0 is feint, S-reference-0 is v06. The duplicated cells are run as named and used as a determinism check.                                                                                                                                                                                                                                                                    |
| <b>D16</b> | Note             | Z08 fits 55 coordinates of which 18 are inert. The protocol specifies <code>--base=v07i</code> for the white-box class; <code>v07i</code> derives from V06Agent and reads only the 37 v0.6 coordinates ( <code>factory.hpp</code> ), while the CEM extends the vector to 55 for any <code>v07*</code> base. The search is over 37 live coordinates plus the inverter the base fixes.                                                                               |
| <b>D17</b> | Note             | thread counts. Every scored cell ran at <code>--threads=13</code> , which section 5.3(2) fixes for the preregistered batteries. The S6 gate condition ran at <code>--threads=1 --freshagents</code> , and the S6 residual column at <code>--threads=1</code> alone.                                                                                                                                                                                                |
| <b>D18</b> | Correction To D8 | a scored cell is NOT bit-reproducible for a searching arm at panel sizes. The B0 check found five runs of a 400-deal cell bit-identical. The panel-duplicate check, on 1,500- and 6,000-deal cells, found one pair of identical commands disagreeing by one game in 12,000 – the cross-deal agent residue of section 5.3 reaching the scored mode. The magnitude is 0.008 points against a 0.63-point half-width. Reported in full under "Panel duplicates" above. |

*continued overleaf*

| ID         | Kind             | Deviation, addition or correction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D19</b> | Correction To D7 | B0.3's changed-source NOTE could not fire, and reporting zero as "conservative" was wrong. <code>freeze_config_v07.py</code> <code>--verify-only</code> reads its drift baseline out of <code>engine/fishbot_v07.json</code> 's own <code>provenance.srcSha256_16</code> . That artifact was REWRITTEN after the freeze commit 0fa4a5f – at d1c6b35 and again at d8c554b – and each rewrite refreshes the baseline to the then-current tree. B0.3 therefore compared the tree against a baseline taken from the...                  |
| <b>D20</b> | Disclosure       | five commits landed in the repository DURING the battery, and four engine sources now differ even from the refreshed baseline: <code>fish.hpp</code> , <code>game.hpp</code> , <code>human.hpp</code> , <code>table.hpp</code> , from the "web play" commits b8cb227, 2e829c2, 46f3515, 72fa936 and e0da8fd. They touch the interactive browser table, not the policy. The decisive fact is that engine/fish7 was NEVER REBUILT: sha256 cf6d5ea2c1f0e9e3896b..., mtime 2026-08-25 13:19, identical at the start of B0 and at the... |
| <b>D21</b> | Ordering         | B0 was performed before anything else, but its ARTIFACT was written later. The five B0 checks were run interactively at the top of the session, before the commit gate and before any holdout deal was played. <code>engine/p5_b0.py</code> , which re-runs all five and records them as P5-B0.json, was written and run about fifteen minutes into B2. Every B0 check is deterministic and reproduced identically, so the artifact records what was checked first – but its file...                                                |
| <b>D22</b> | Observation      | one action-limit hit in 428 scored cells: the phase-2 composite against SEALED:S-ask-1 on bank 7090001, one game in 12,000 (limitHitRate 8.33e-05). G4 gates the mirror pathology run and not scored cells, so this is not a gate violation, but the corpus standard is zero and it is recorded. NOTE that scored cells carry NO audit evidence: <code>auditChecks</code> is 0 in all 428 of them, because <code>match</code> runs the audit only under <code>--audit</code> and no preregistered cell...                           |
| <b>D23</b> | Correction       | the throughput columns were printed with <code>median_high(v[len(v)//2]</code> on an even-length list) rather than the median. Corrected to the true median; every figure moves by at most 0.11x and every one still exceeds the 3.2x the protocol records.                                                                                                                                                                                                                                                                         |
| <b>D25</b> | Reading          | section 6 item 3 read per bank. The protocol states the one-third location rule without a per-bank clause, and section 3's replication rule is written for "a claim whose SIGN does not replicate". Here the sign of the drop does not change between banks; the VERDICT LABEL does – LOCATED on 7090001, NOT LOCATED on 7090003. Applying section 3's rule to a threshold crossing rather than to a sign is an extension, and section 7 requires a deviation for...                                                                |

*continued overleaf*

| ID  | Kind           | Deviation, addition or correction                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D26 | Substitution   | B11’s third clause is not computable as written. It asks phase 5 to confirm “that the holdout estimate is not systematically smaller than the training estimate by about that amount”, where the estimate B11 is about is the measured holdout gain – but the protocol supplies no training figure for B2.1 or B2.2. Phase 5 substituted the three quantities the protocol DOES state in advance (the B2.4 group at +0.78, the B6 self-play row at +2.94, and B7’s...                       |
| D27 | Added Artifact | <code>research/v07/results/P5-drift.json</code> . D19’s executed replacement for the vacuous hash-based drift check was performed but not initially recorded, which made it an assertion rather than evidence. It is now an artifact: the build command, both binaries’ SHA-256 and byte counts, the pathology argy, the two recomputed digests, and the source sets differing from the artifact’s own baseline and from the freeze commit.                                                 |
| D24 | Correction     | per-bank columns were printed unlabelled and sorted by seed NUMBER, which inverts the attribution for B5 and B8, whose primary bank is 7090003 and whose replicate is 7090001. Every per-bank column in <code>P5-TABLES.txt</code> is now printed as <code>seed:value</code> . In THIS document the bank order is given in each table’s first row and the pairs that follow keep it: section 5’s B4 table is (7090001, 7090002) and section 9’s B8 table is (7090003, 7090001). A reader... |

## G Reproduction and availability

### Availability

The engine, every experiment artifact, the protocol, the generator that produces this paper’s numbers and the provenance checker that audits them are in the project repository, <https://github.com/dylann29/fish-optimization>. The evaluation of §11 ran at the commit whose current hash is 4d8cec9. Two caveats attach to that hash and the subsections below give both: the repository’s history was rewritten after the evaluation, and no single commit describes the tree for the whole run in any case.

### A history rewrite, and what it did to the hashes

After the evaluation was complete, one artifact of the development phase, a per-decision search capture of 139,267,276 bytes, exceeded the hosting limit and was excised from version-control history so the repository could be published. That rewrote every commit from phase 3 onward.

Three consequences, stated because a reader checking this paper against the repository will otherwise be misled. First, **the commit hashes recorded inside the phase-5 artifacts no longer resolve**: the verification block records d920f5b and the freeze artifact’s drift check keys on 0fa4a5f, and neither object exists in the published history. Every hash printed in this paper is the post-rewrite one, recovered by matching commit subject lines, so the freeze commit is ed321cc and the evaluation commit is 4d8cec9. Second, the seal predates the rewrite and is unaffected: the commit at which the holdout material was sealed, f4581da, still resolves, so the commitment chain of §7.6 is intact and independently checkable. Third, the excised file is retained on disk and

its digest recorded beside it in `research/v07/results/K5-capture-7030004.sha256`, SHA-256 `bb655aac7cd413d2...`; it is regenerable from the capture harness and no measurement in this paper depends on it.

The rewrite is also the clearest possible argument for the subsection that follows.

## The run is identified by its binary, not by a commit

Five commits landed in the repository while the battery was running, so no single commit describes the tree for the whole of it, and a rewrite has since changed all of their hashes. Every one of the 428 scored cells was played by `engine/fish7`, SHA-256 `cf6d5ea2c1f0e9e3896bc51f2a366834e7d60303eaf85acc553346c97359f4e3`, 1,242,312 bytes, and that binary was never rebuilt between the first bank digest and the last cell. The five later commits touch the interactive browser table and not the policy; §E.2 records the executed check, rebuilding the engine from the tree as it stood at the end of the battery and recomputing the mirror digest, that establishes the frozen policy is unmoved by them.

## The battery

Table 26 repeats Table 7.

**Table 26:** Every registered battery against what was measured. *Registered* is the count the protocol names, not the count that ran: deviations D1 and D2 add a B5 reference cell and sixteen B9 cells, because without them “no single leave-one-out drop exceeds the whole”, “the recovered size tracks the planted size” and “the planted excess resolves to zero” are not computable from the cells the protocol lists. No cell was dropped.

| Battery | What it measures                       | Registered | Added | Measured | Status        |
|---------|----------------------------------------|------------|-------|----------|---------------|
| B2      | headline strength against the frontier | 10         | 0     | 10       | complete      |
| B3      | the shared 31-member panel             | 248        | 0     | 248      | complete      |
| B4      | a fresh adversary search               | 16         | 0     | 16       | complete      |
| B5      | the attribution lattice                | 22         | 2     | 24       | complete (D1) |
| B6      | the partner-regime table               | 64         | 0     | 64       | complete      |
| B7      | cross-play between independent runs    | 24         | 0     | 24       | complete      |
| B8      | the rule-dialect table                 | 16         | 0     | 16       | complete      |
| B9      | the negative controls                  | 10         | 16    | 26       | complete (D2) |
| B10     | the S6 residual                        | 16         | 0     | 16       | complete      |

## Where the raw output is

Under `research/v07/results/`, digested in `research/v07/results/MANIFEST-P5.json`: 29 artifacts, of which 29 re-verify against their recorded SHA-256 and byte count on disk at the time this document was generated.

- `P5-B0.json`, the verification block: seven bank digests, the sealed plaintext hash, the freeze round-trips, the seed registry, the engine self-audit, and the reproducibility runs.
- `P5-gate.jsonl`, the commit gate for all four configurations, rule by rule.
- `P5-B2.jsonl ... P5-B9.jsonl`, one JSON row per cell, each carrying the literal argument vector it was produced by and the complete match object the harness emitted.
- `P5-B10.jsonl`, one row per audited cell. These are seat-isolation audits rather than matches and carry a `side` object with the mismatch count and its denominator, not a match object.



- `P5-Z01.jsonl ... P5-Z08.jsonl`, `P5-B9-h*.jsonl`, the fitting traces: header, one record per generation, and the final weight vector each adversary spec was rebuilt from.
- `P5-TABLES.txt`, the phase-5 reduction, produced by `engine/p5_analyse.py`.

Every row carries its own argument vector, so any single cell can be re-run by hand from the artifact without reference to this document or to the driver that produced it.

## How the numbers in this document were produced

1. `engine/build_tables_v07.py --paper` reads the artifacts above and writes `paper/numbers_v07_generated.tex` and `paper/tables_v07/*.tex`. Every macro it emits carries a comment naming the artifact and field it came from, and re-emitting a name with a different value raises rather than silently winning.
2. `paper/check_provenance.py --version v07` audits the result: a number used in a section must either be generated from an artifact or sit under a comment header naming an artifact that exists on disk. Of the numbers the sections use, 250 are emitted by this cycle's generator, all but the registered training figures of Table 10 computed from the phase-5 artifacts and those four carried in the generator as constants of the protocol; 24 are generated from earlier cycles' artifacts by those cycles' generators; and 146 are transcribed under a header naming a source document.
3. `paper/inline.py` flattens the modular source into a single self-contained file.

Sections 2–4 quote results from this programme's earlier cycles. Those numbers are not retyped here either: the earlier cycles' generators emitted them from their own artifacts, and this document loads those generated files directly. The split is therefore 250 generated by this cycle, 24 generated by an earlier one, and 146 transcribed under a header naming the report or artifact that records them, which `paper/check_provenance.py` verifies exists on disk, and whose printed split this paragraph is required to match.

The generated file is `\input` after the placeholder file, so a value in the placeholder can never reach a PDF built after the battery has been run. Generated on 2026-08-30.

## Sample sizes

The project's power rule is a 95% half-width of  $98/\sqrt{N}$  pp at parity, computed by the harness and emitted with every cell. At 48,000 games the primary claim's pooled half-width is 0.45 pp; the frontier cells are sized so that the primary claim's half-width is small against the 1.53-point reference bar rather than against the effect. Far-class panel members are sized at a quarter of the near class deliberately: they are decided by 24.72 to 50.00 pp and do not need the resolution.